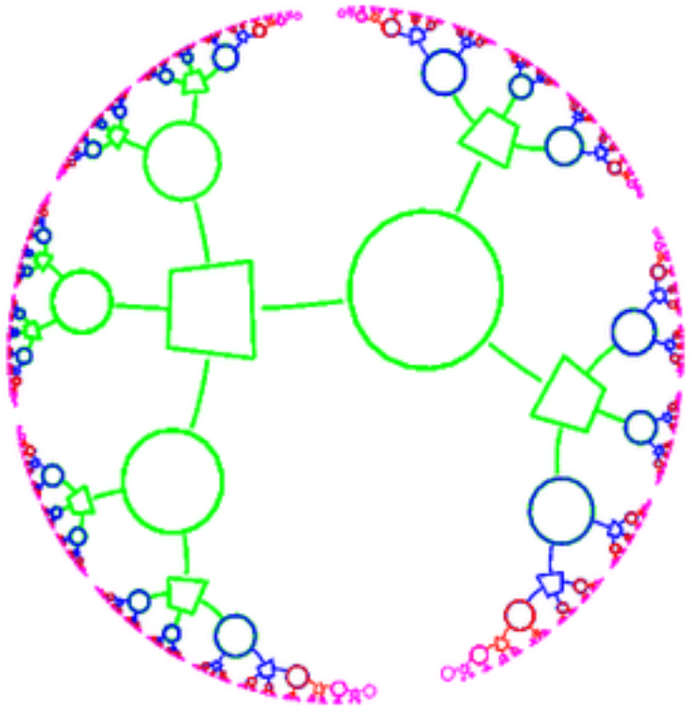
Monte Carlo Methods

— Part 1 :

Simple methods

& Basic Markov Chain Monte Carlo

(MCMC)



Information theory, pattern recognition, and neural networks

- 1 Noisy-channel coding

Source coding (Data compression)

- 2 Information content, entropy
- 3 Typicality and the source coding theorem
- 4 Symbol codes
- 5 Symbol codes and Arithmetic coding

Noisy-channel coding

- 6 Information measures for noisy channels
- 7 Capacity of a noisy channel
- 8 The noisy-channel coding theorem

Inference + probabilistic methods

- 9-10 Inference
- 11 Clustering
- 12-13 Monte Carlo methods
- 14 Variational methods

www.inference.phy.cam.ac.uk/itprnn/

www.inference.phy.cam.ac.uk/itila/

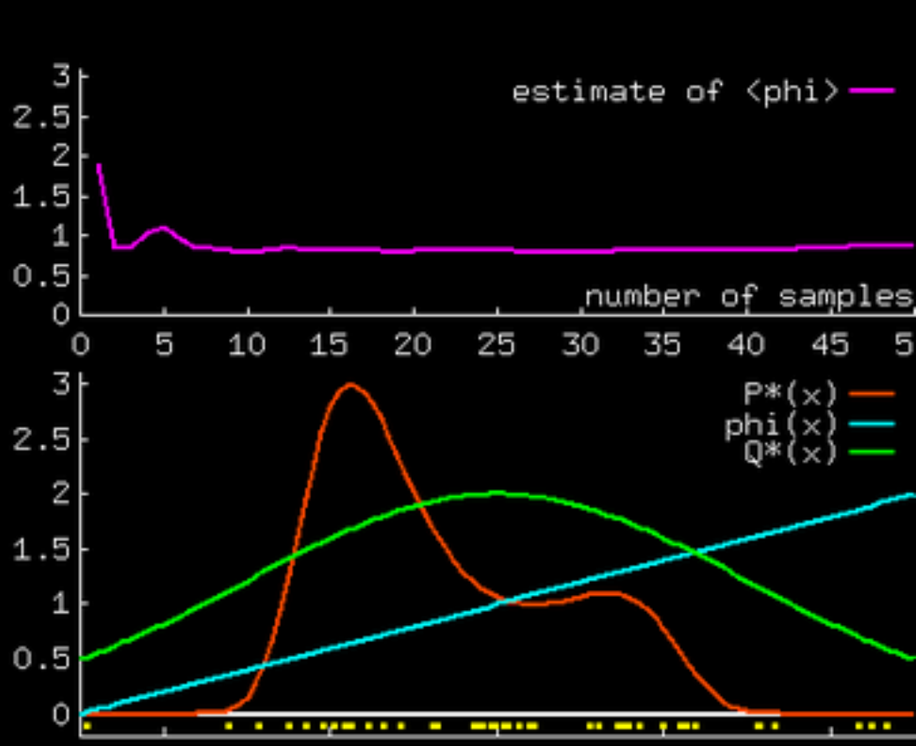
Probability tool box

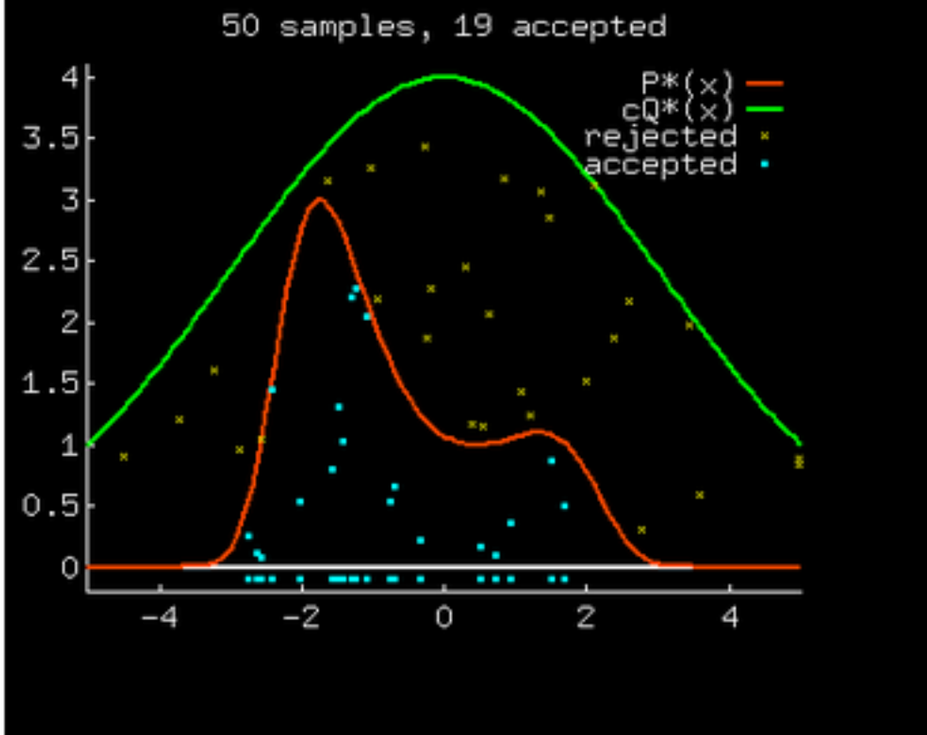
- Complete enumeration
- Laplace's method
- Monte Carlo methods
- Variational methods

Monte Carlo methods

- Simple Monte Carlo methods
  - Importance sampling
  - Rejection sampling
- Markov-chain Monte Carlo methods
  - Metropolis method
  - Gibbs sampling
  - Slice sampling
- Reducing random-walk behaviour
  - Hamiltonian Monte Carlo
  - Overrelaxation
- Exact sampling

itp/exact/rc RUNME





Lecture 11

Clustering

Lecture 12

Monte Carlo methods

Lecture 13

Advanced Monte Carlo methods

Monte Carlo methods

- Simple Monte Carlo methods
  - Importance sampling
  - Rejection sampling
- Markov-chain Monte Carlo methods
  - Metropolis method
  - Gibbs sampling

  - Slice sampling
- Reducing random-walk behaviour
  - Hamiltonian Monte Carlo
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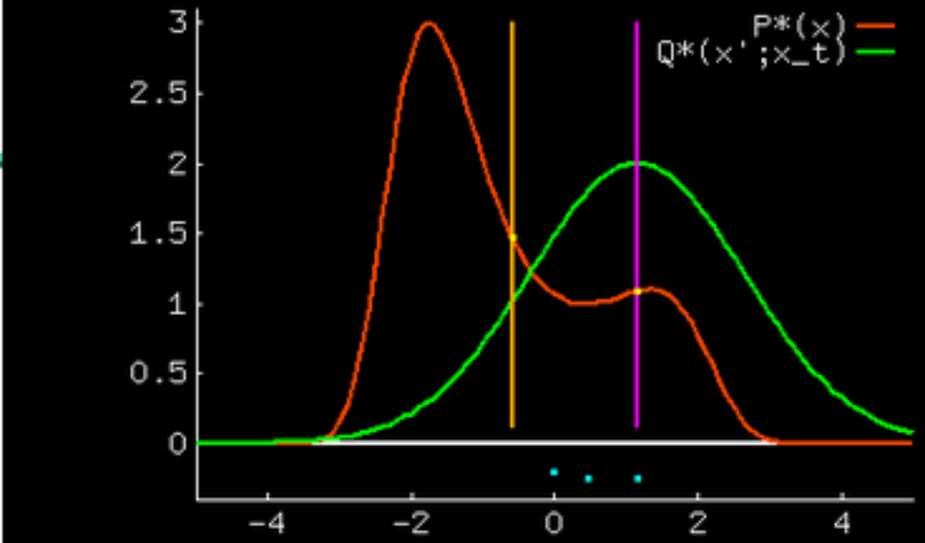
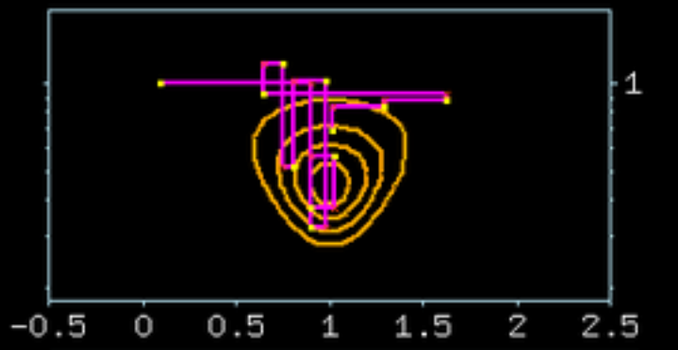
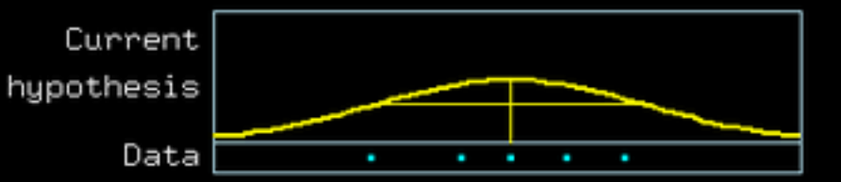
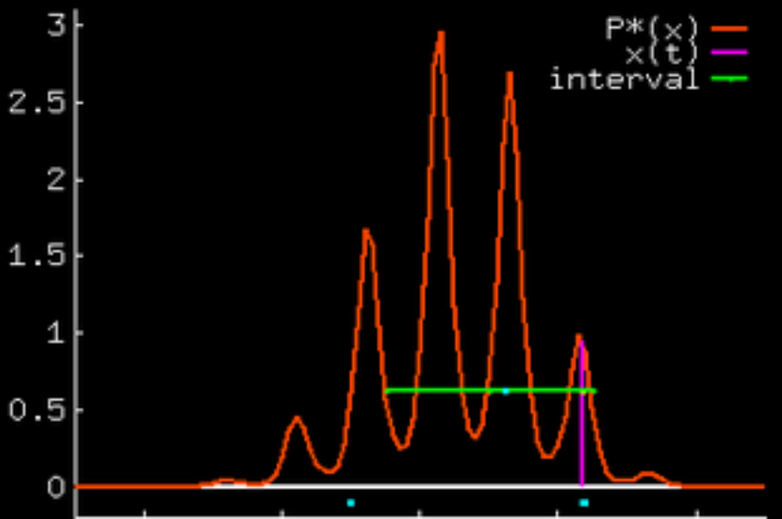
itp/exact/rc RUNME

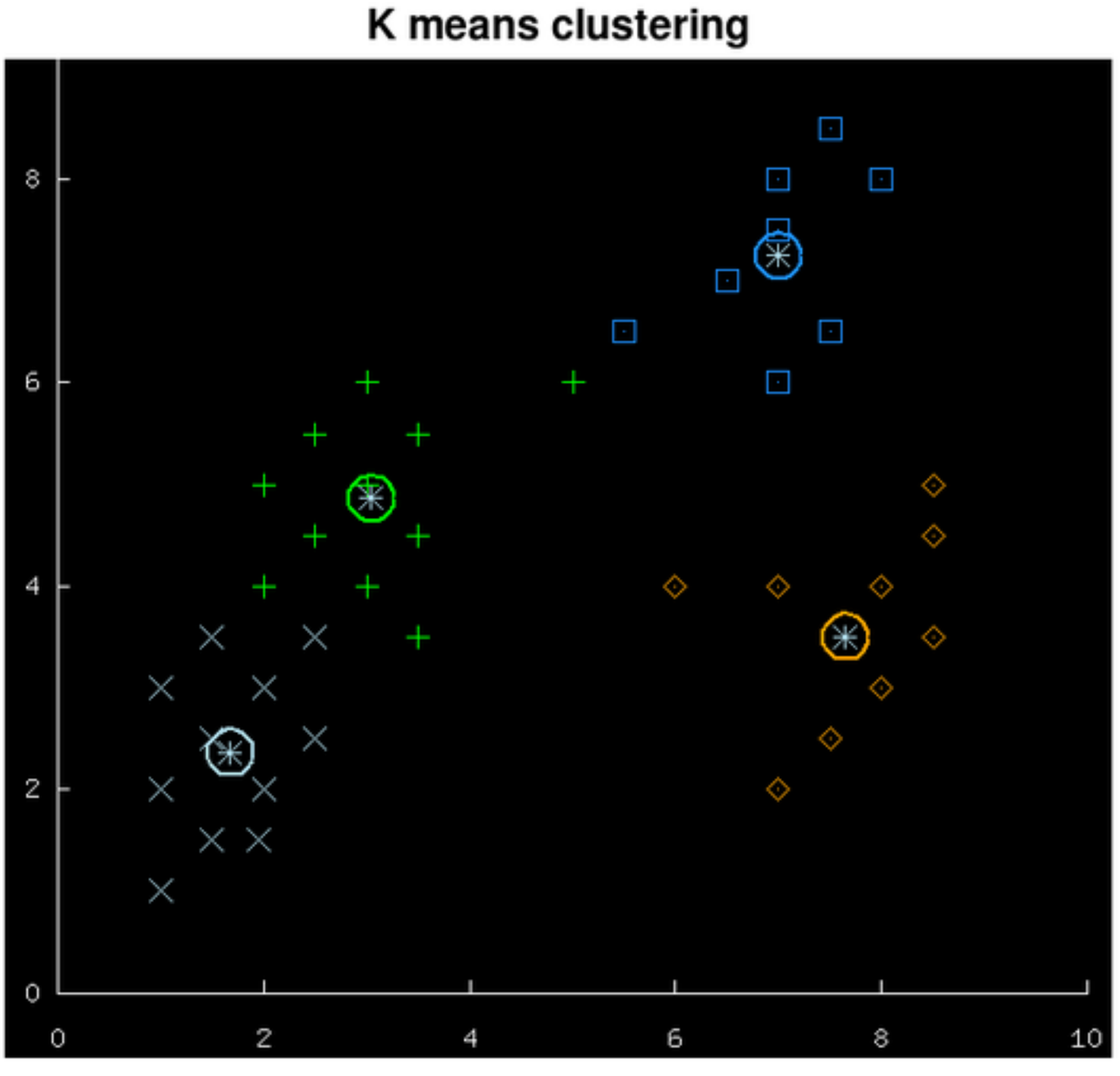








Clustering



Posterior probability of

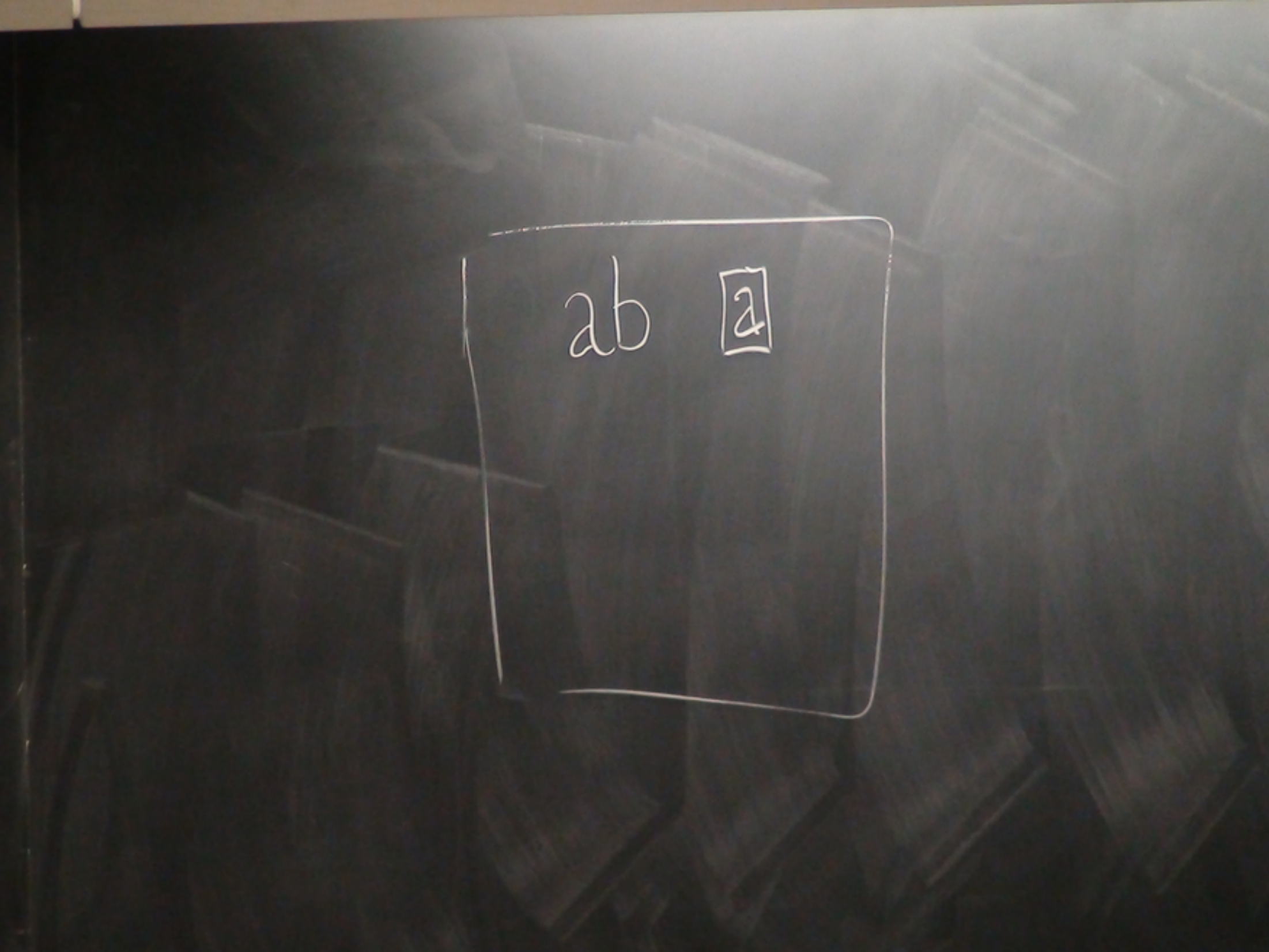
$\underline{\mu}_1, \underline{\mu}_2 \quad \underline{\sigma}_1, \underline{\sigma}_2 \ldots, \pi_1, \pi_2 \ldots \pi_k, \quad C_1, C_2, C_3 \ldots C_N$,

$C_1 \in \{1, 2, 3 \ldots K\}$

given Data $\underline{X}_1, \underline{X}_2 \ldots \underline{X}_N$

& given the assumption that

$\underline{X} \sim$ mixture of T Gaussians


ab
a

Posterior θ probability of

$\underbrace{\mu_1, \mu_2 \ldots \quad \sigma_1, \sigma_2 \ldots, \pi_1, \pi_2 \ldots \pi_k}_{\theta}$, $\underbrace{c_1, c_2, c_3 \ldots c_N}_{c}$,

$c_1 \in \{1, 2, 3 \ldots K\}$

&

$$P(\theta, c \mid \{x\}) = \frac{\prod_1^N P(\{x\} \mid \theta, c)\, P(c \mid \theta)\, P(\theta)}{P(\{x\})}$$

$$\log P(\theta, c \mid \{x\}) \rightarrow \sum_1^N \log \pi + \frac{(x-\mu)^2}{\sigma^2} + \log\left(\sqrt{2\pi\sigma^2}\right).$$

$$\log P(\{x\} \mid \theta, c)$$

$$P(\theta, c \mid \xi \times \xi) = \frac{P(\xi \times \xi \mid \theta, c)\, P(c \mid \theta)\, P(\theta)}{P(\xi \times \xi)}$$

$$P(\theta, c \mid \xi \times \xi)$$

$$\log P(\xi \times \xi \mid \theta, c) + \log P(c \mid \theta)$$

$$\sum_{1}^{N} \log \pi_{c_n} + \frac{(x-\mu)^2}{\sigma^2} + 1$$

$$= \frac{\prod_{i}^{N} P(x_i, y_i \mid \theta, c)\, P(c \mid \theta)\, P(\theta)}{P(x, y)}$$

$$\sum_{c} \int \int \cdots \int d\mu_1 d\mu_2 d\sigma \, d\pi \ldots$$

$$\rightarrow \sum_{1}^{N} \log \pi_{c_n} + \frac{(x-\mu)^2}{\sigma^2} + \log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right).$$

Rules of the game

Example

Posterior θ P

$$P(x) = \frac{P^*(x)}{Z}$$

x: the unknown variables

μ_1, μ_2 σ_1, σ

① can evaluate P^*

② don't know Z, but it is of interest

Monte Carlo methods

Interested in $P(\mathbf{x}) = \frac{1}{Z} P^*(\mathbf{x}) = \frac{1}{Z} e^{-E(\mathbf{x})}$.

Can evaluate $E(\mathbf{x})$ at any $\mathbf{x}$.

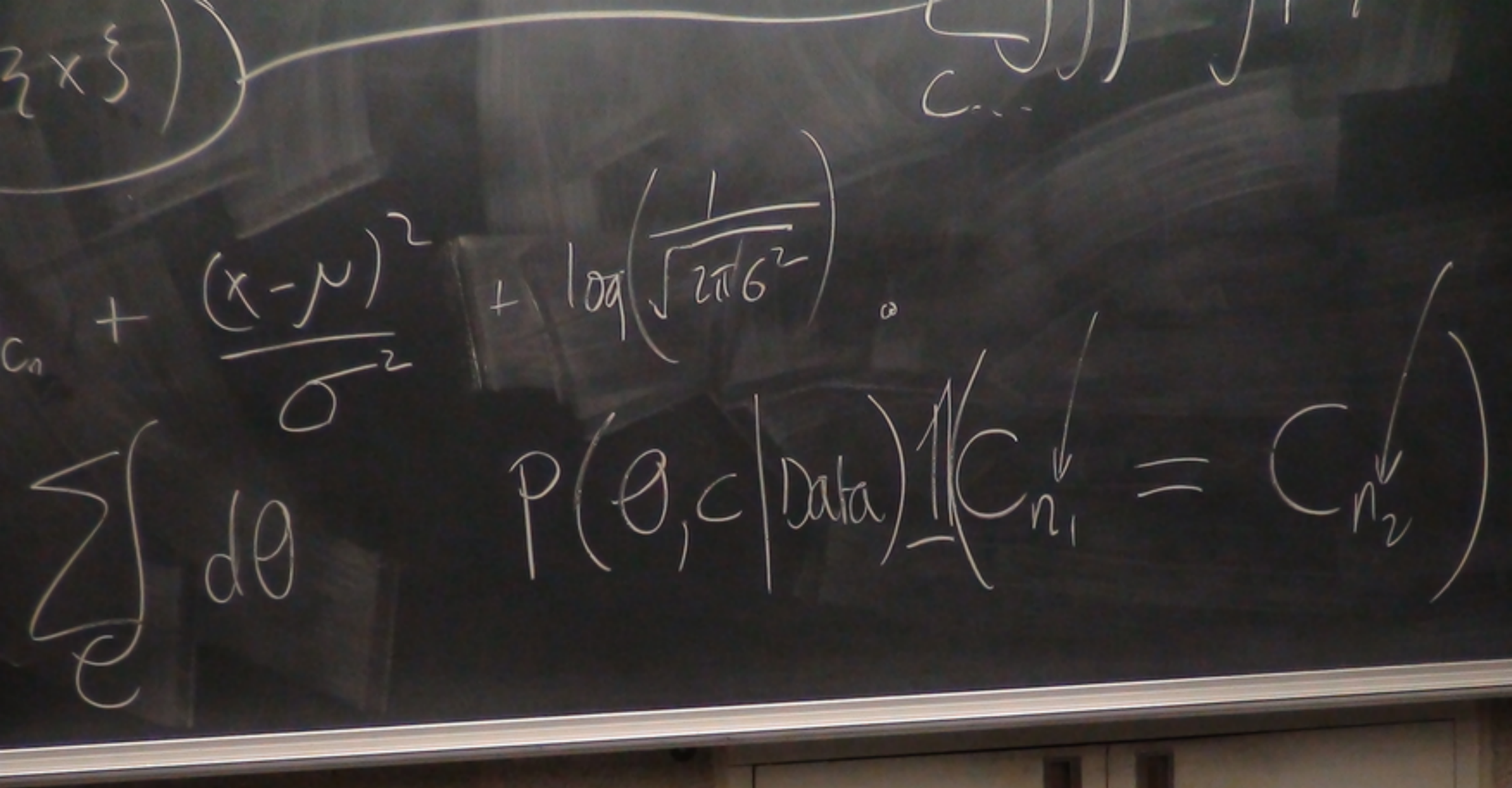
Problem 1 Make samples $\mathbf{x}^{(r)} \sim P$

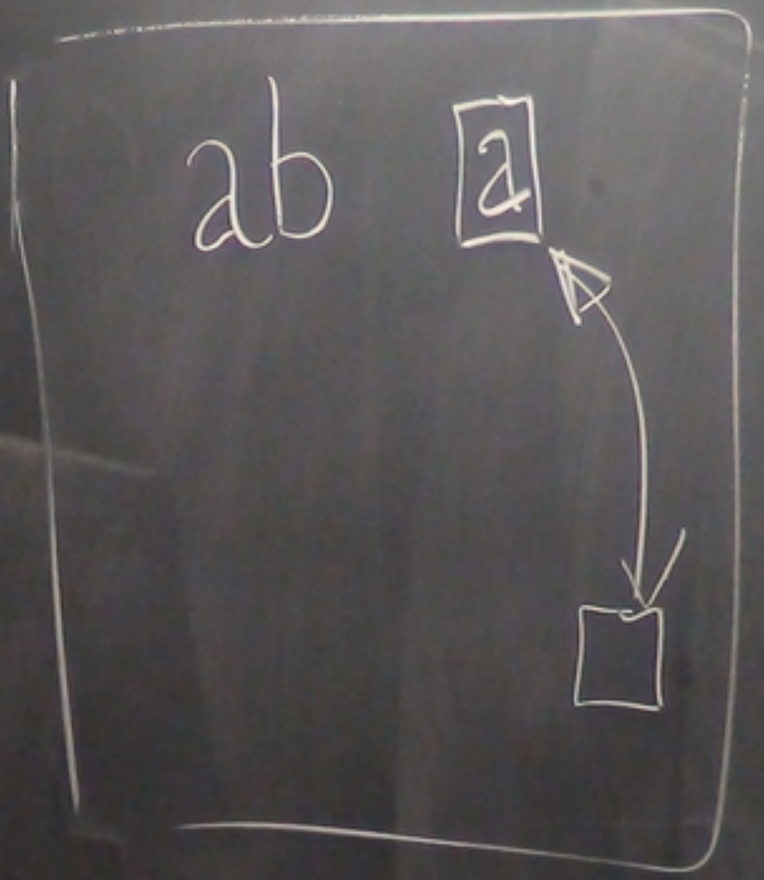
Problem 2 Estimate expectations $\Phi \equiv \sum_{\mathbf{x}} \phi(\mathbf{x}) P(\mathbf{x})$

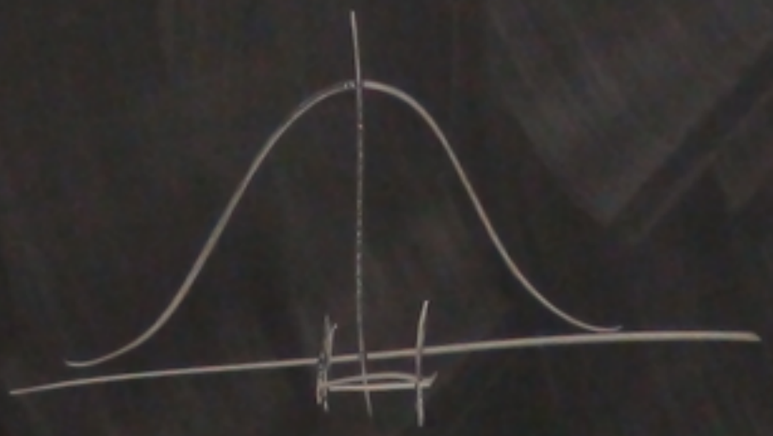
What we're *not* trying to do:

We're not trying to find the most probable state.

We're not trying to visit *all* typical states.


$\frac{(x-\mu)^2}{\sigma^2} + \log\left(\sqrt{2\pi\sigma^2}\right)$
$\int d\theta \; P(\theta, c \mid \text{Data}) \mathbb{1}(c_{n_1} = c_{n_2})$


ab
a



$10 \times 10 \times 10$ spins

32×32 spins

$\underline{x} = S$

$\underline{x} \in \{-1, 1\}^N$

$E(\underline{x})$

$P(\underline{x} \mid \beta, \underline{\underline{J}}) = \frac{1}{Z(\beta, \underline{\underline{J}})} e^{-\beta E(\underline{x}; \underline{\underline{J}})}$

$$E(\underline{x}; \underline{\underline{J}}) = -\sum_{n,n'} J_{nn'} x_n x_{n'}$$

$$\langle E(x) \rangle_P = \sum_{\underline{x}} E(x) P(x)$$

$$2^N = 2^{1000}$$

10^{80} electrons $= 2^{266}$ electrons

age of universe $= \frac{2^{98}\ ps}{2^{364}}$

$\langle m(x) \rangle_P$ $\langle m(x)^2 \rangle_P$ $S = \left\langle \log \frac{1}{P(x)} \right\rangle$

$\log Z \sim$ Free Energy

$P(x)$

10^{80} electrons $= 2^{266}$ electrons

$= 2^{1000}$

age of universe $= \frac{2^{98}}{2^{364}}$ ps

Prob 1

Prob 2

Prob 2 can be solved if Prob 1 solved

$$\{x^{(r)}\}_{r=1}^{R}$$

st

Prob 2 can be solved if Prob 1 solved

If $\{x^{(r)}\}_{r=1}^{R} \sim P(x)$

Can estimate $F = \langle f(x) \rangle_P$ $\quad \hat{F} = \frac{\sum_r f(x^{(r)})}{R}$

Monte Carlo methods

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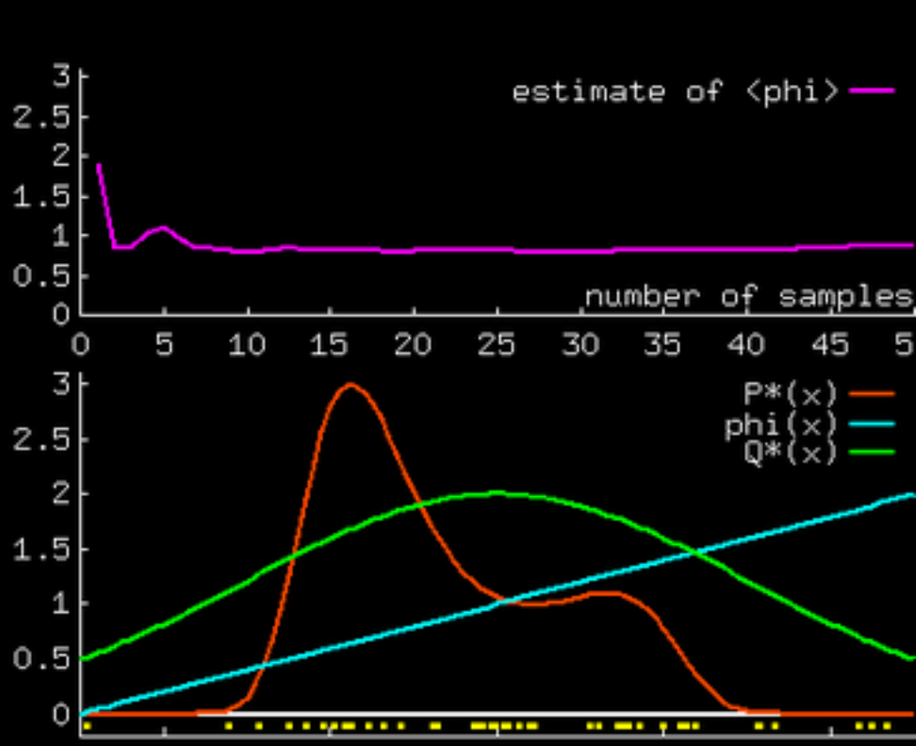
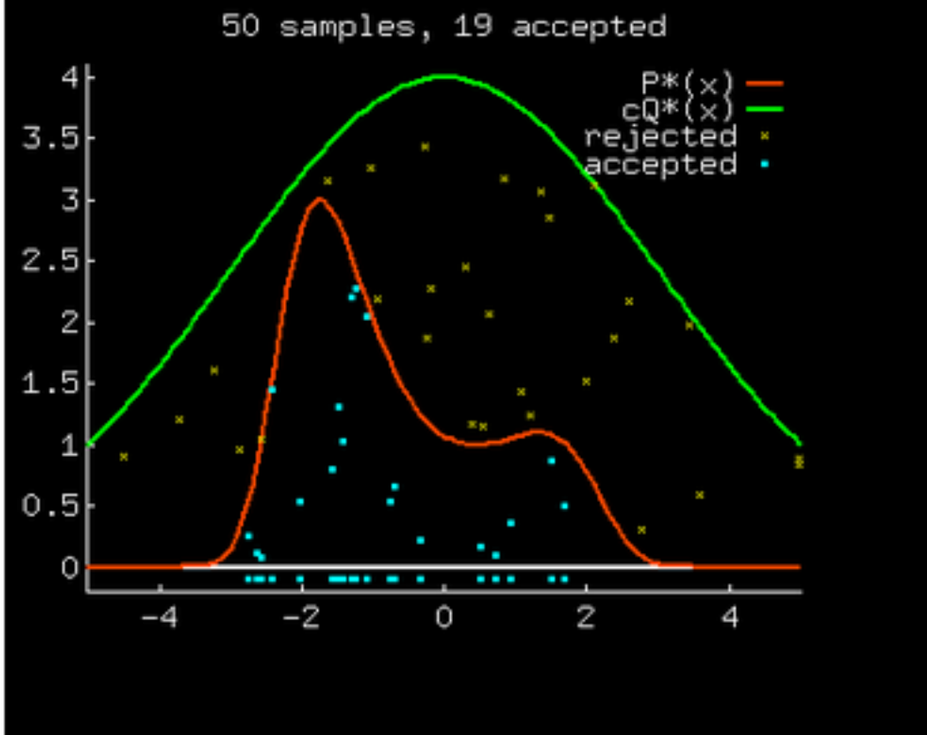
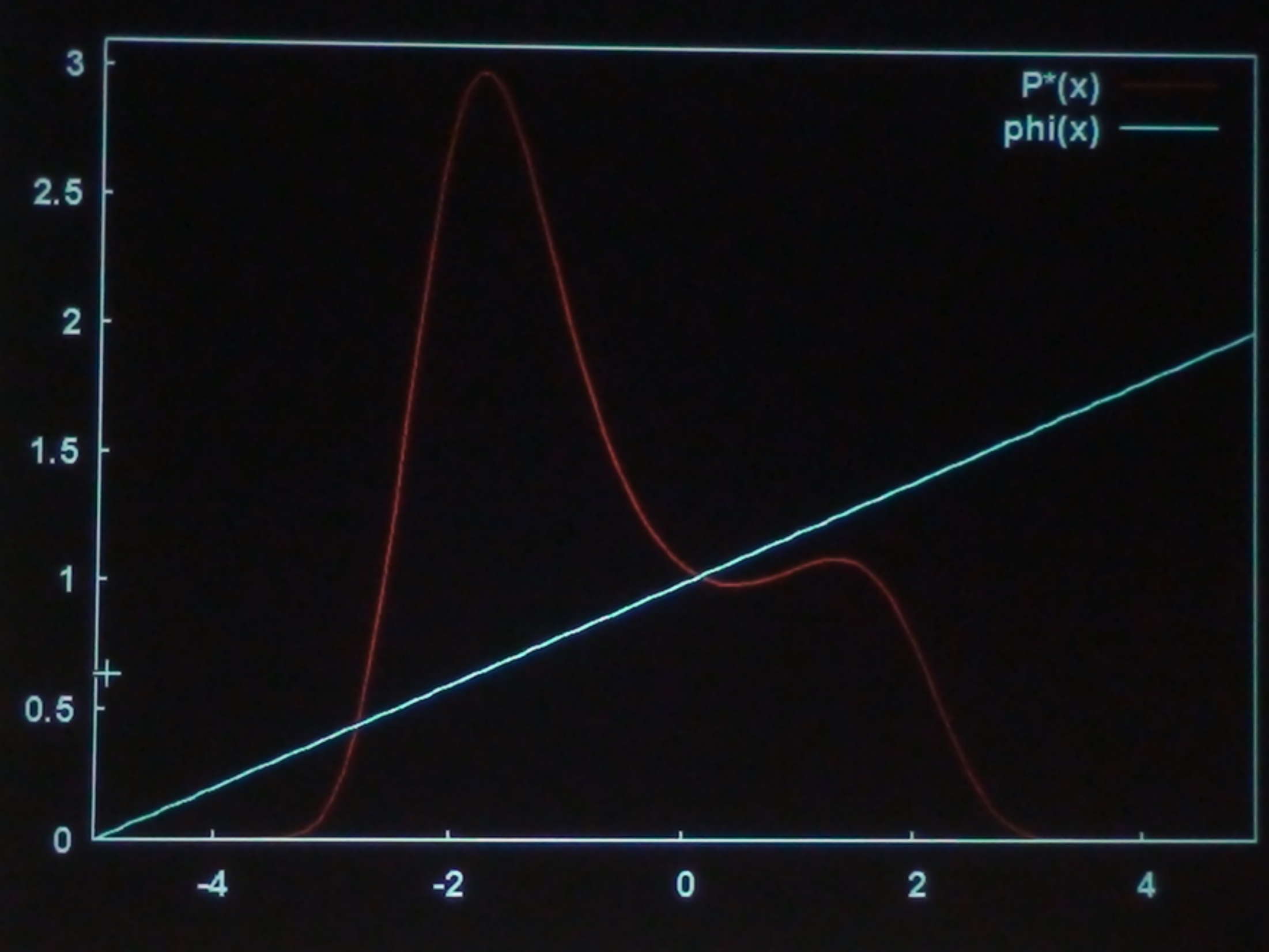
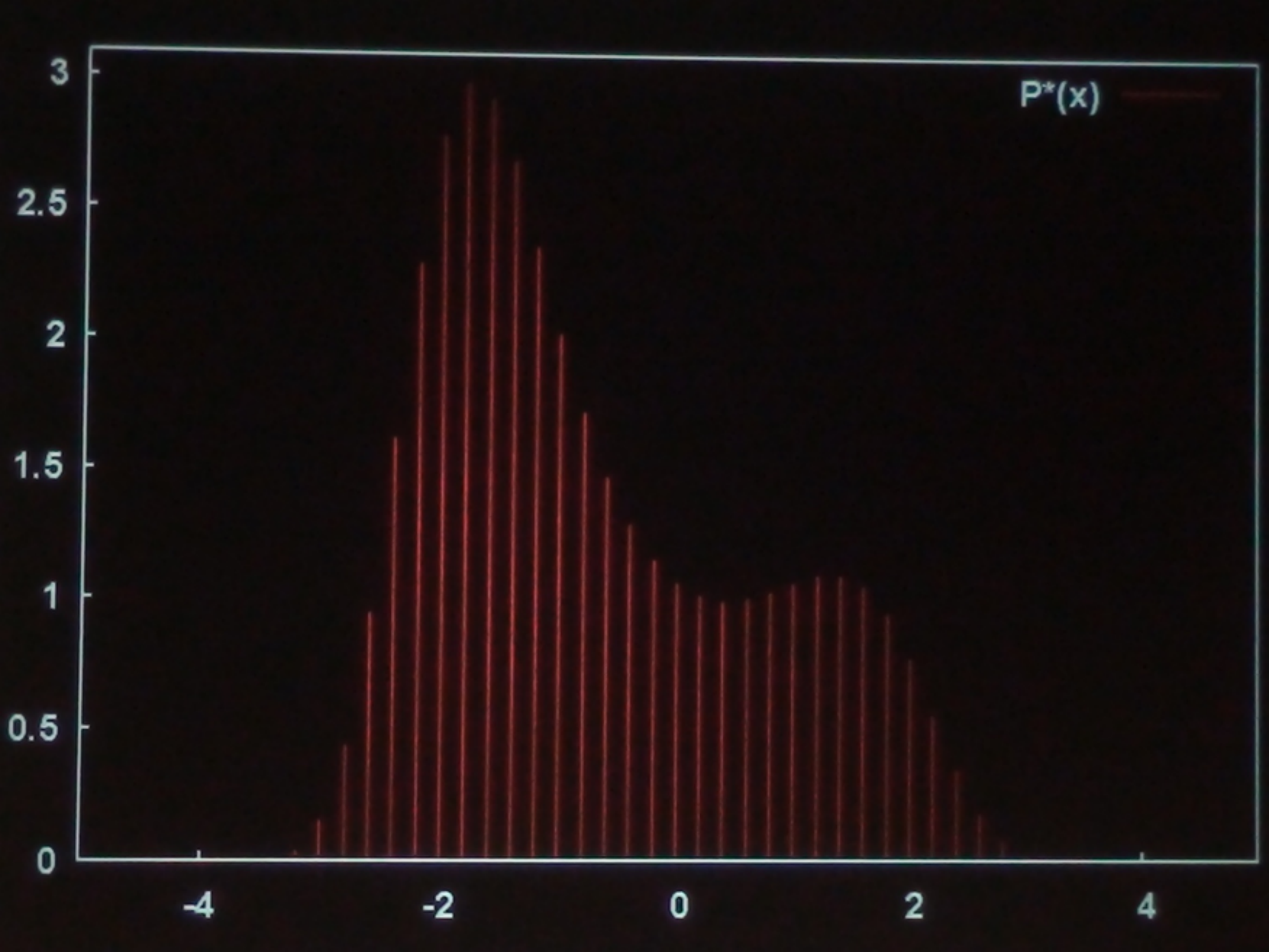
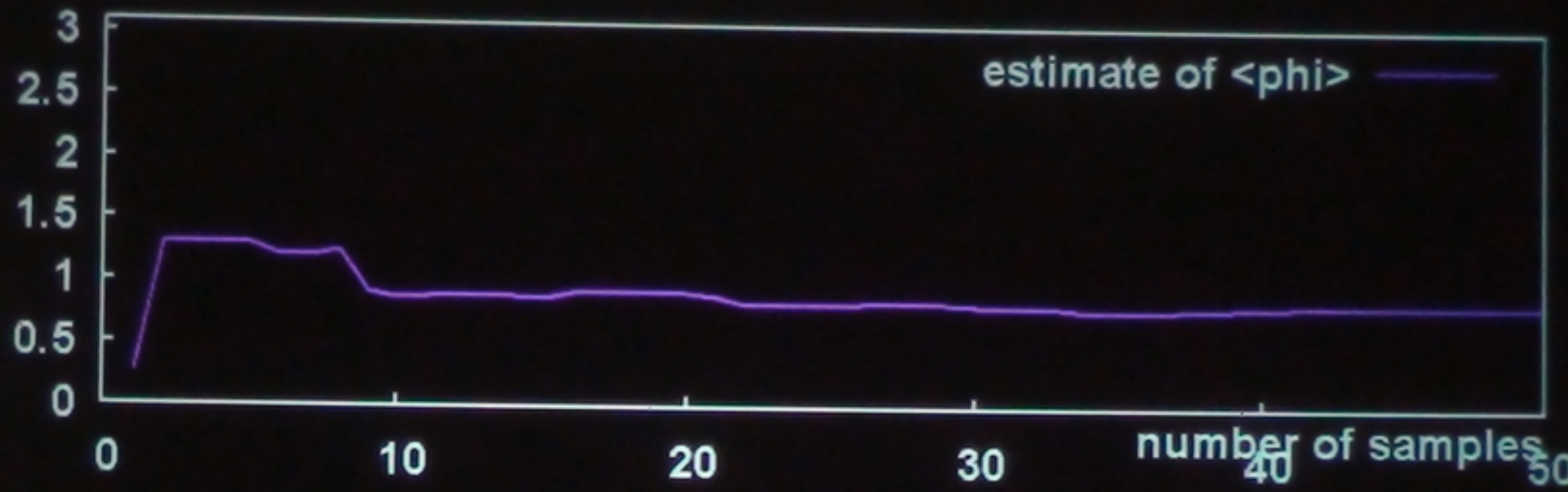
itp/exact/rc RUNME

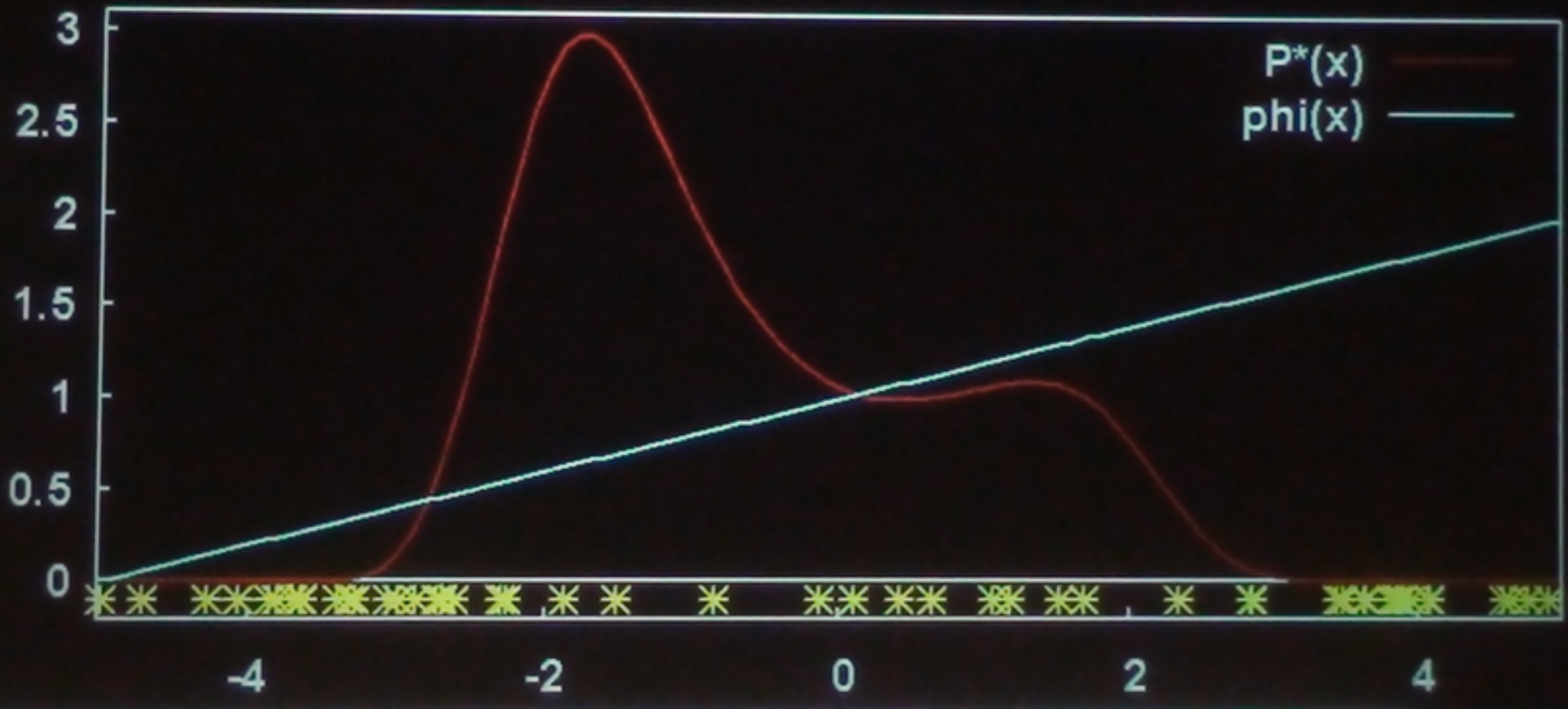


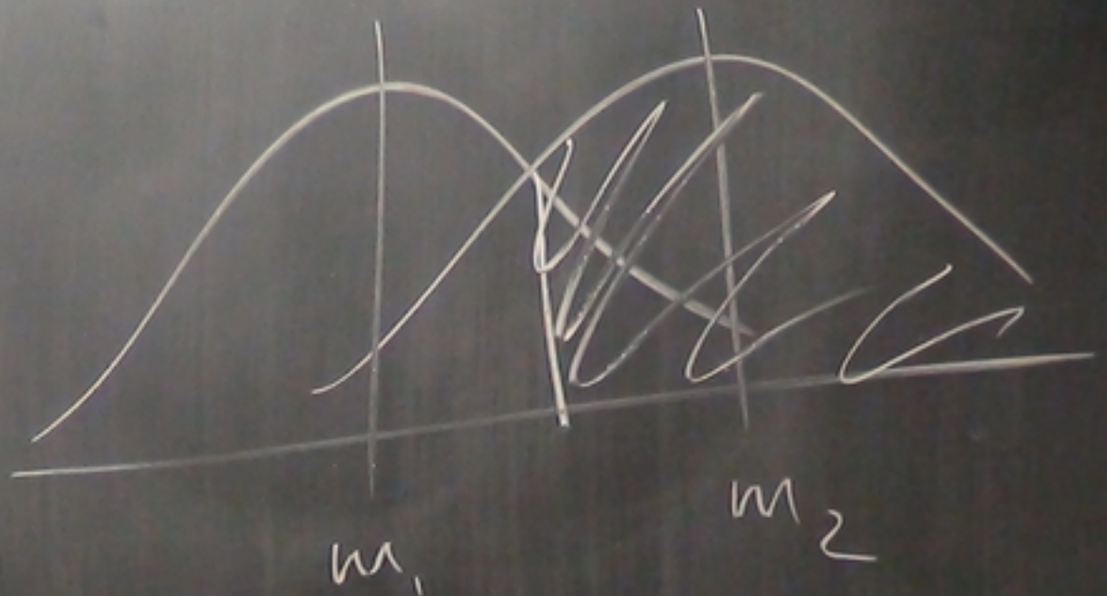


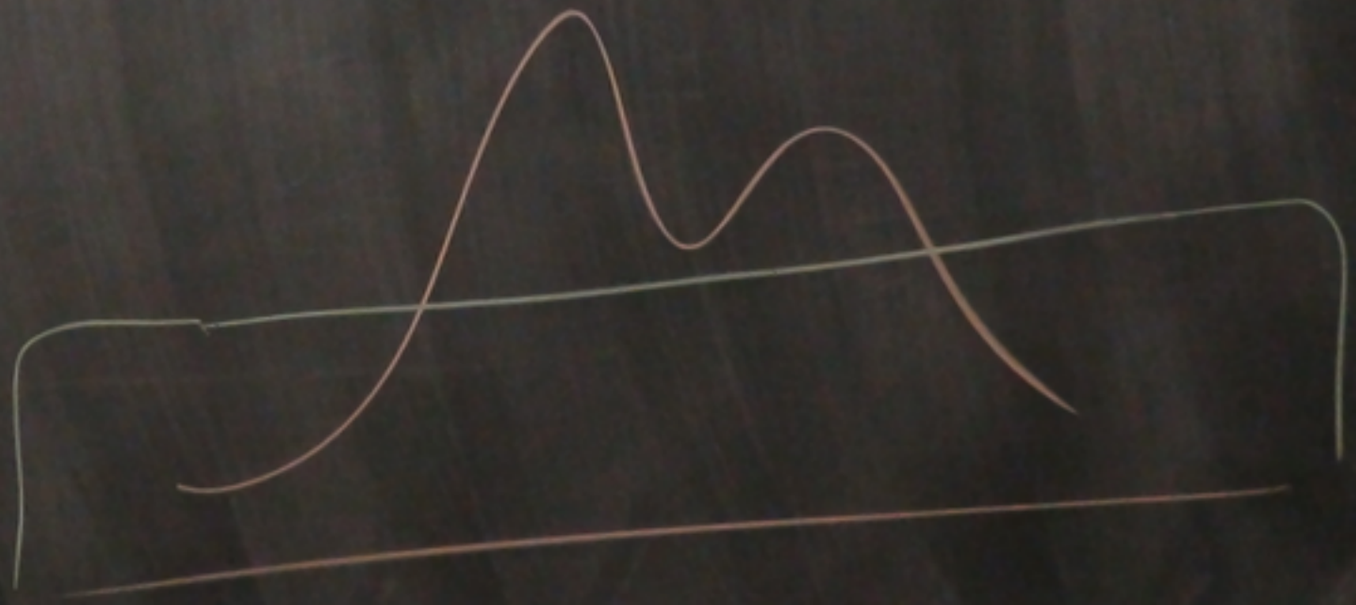

P*(x)
phi(x)
3
2.5
2
1.5
1
0.5
0
-4
-2
0
2
4


P*(x)
3
2.5
2
1.5
1
0.5
0
-4
-2
0
2
4


estimate of <phi>
3
2.5
2
1.5
1
0.5
0
0
10
20
30
40
50
number of samples


P*(x)
phi(x)
3
2.5
2
1.5
1
0.5
0
-4
-2
0
2
4


m_1
m_2



sampler density

can draw from Q

Q = uniform

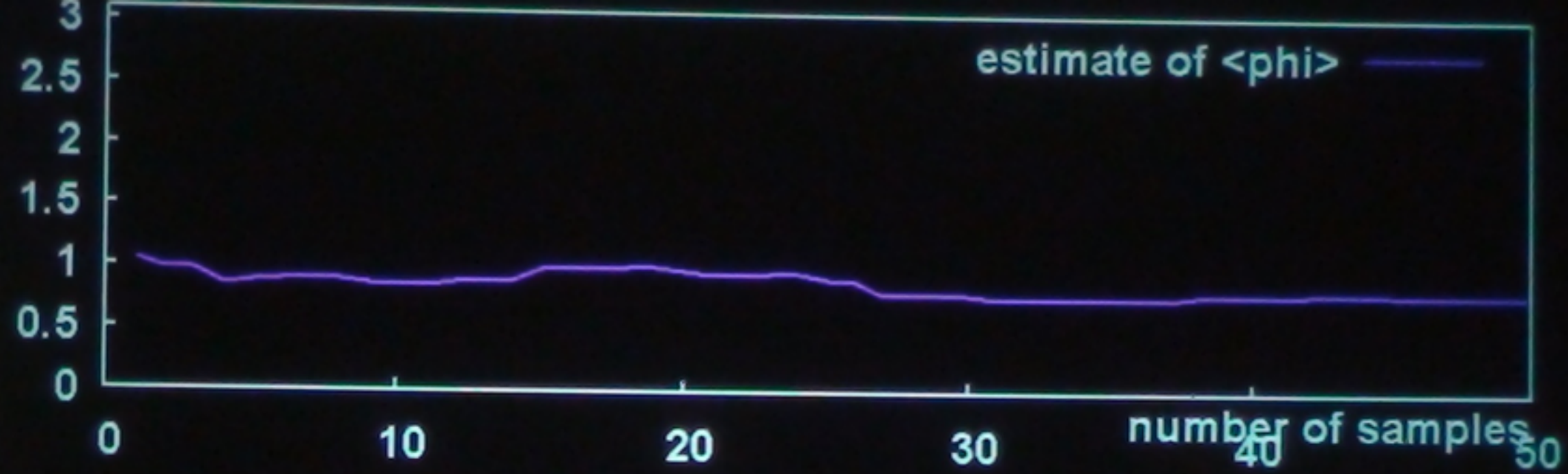
$$\hat{\Phi} = \frac{\sum_{1}^{R} \frac{P^*(x^{(r)})}{Q(x^{(r)})} \phi(x^{(r)})}{\sum_{1}^{R} \frac{P^*(x^{(r)})}{Q(x^{(r)})}}$$

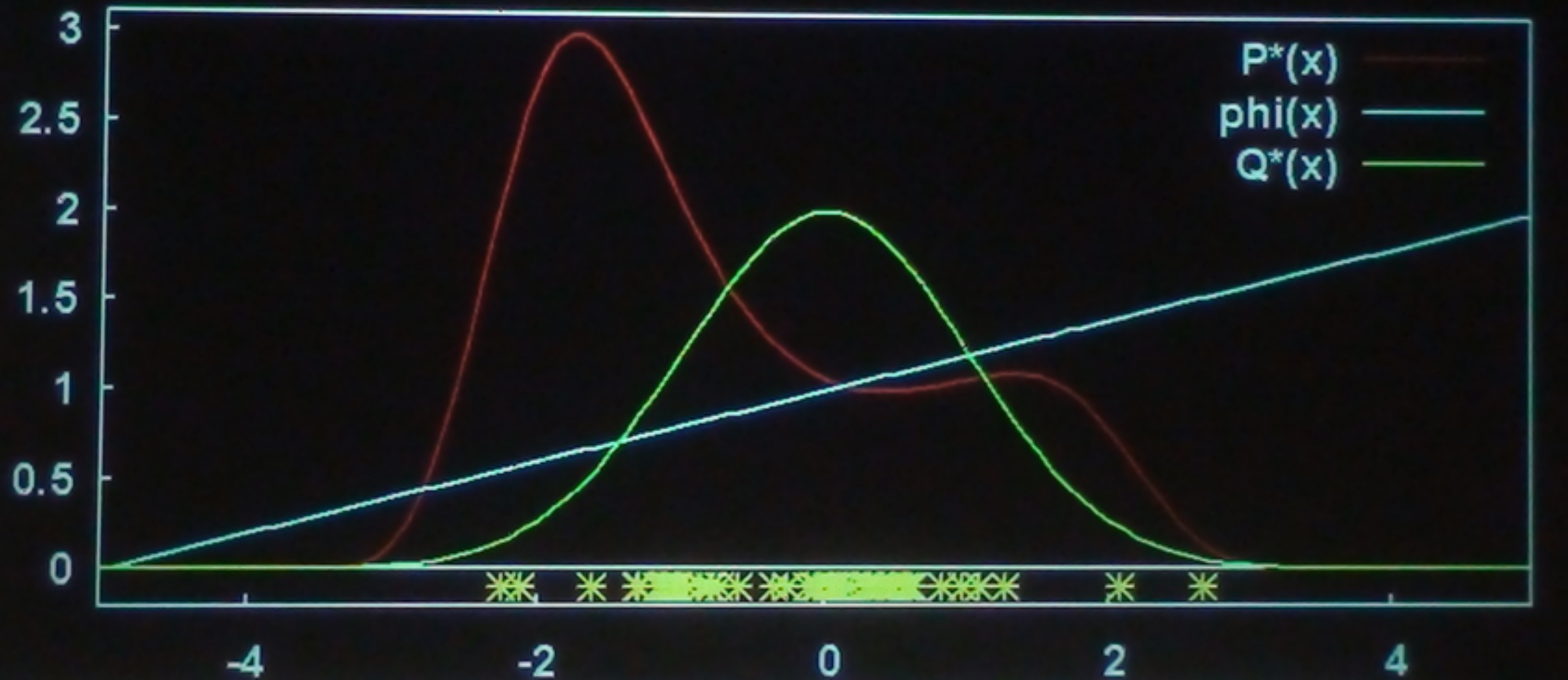
IMPORTANCE SAMPLING

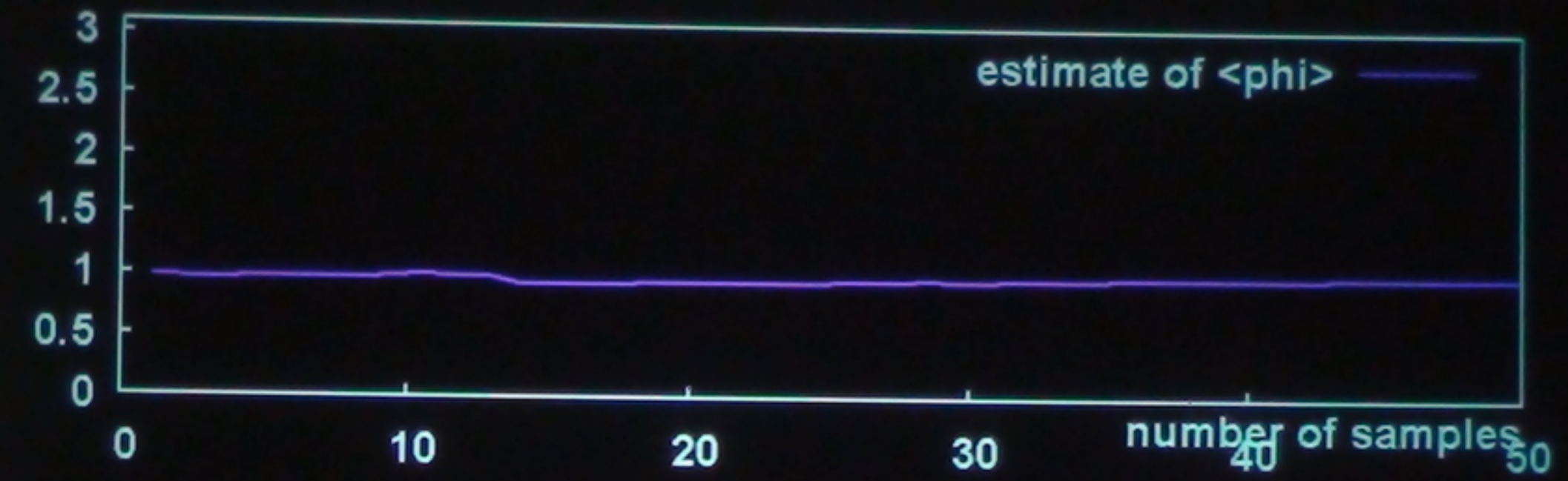
expected value of

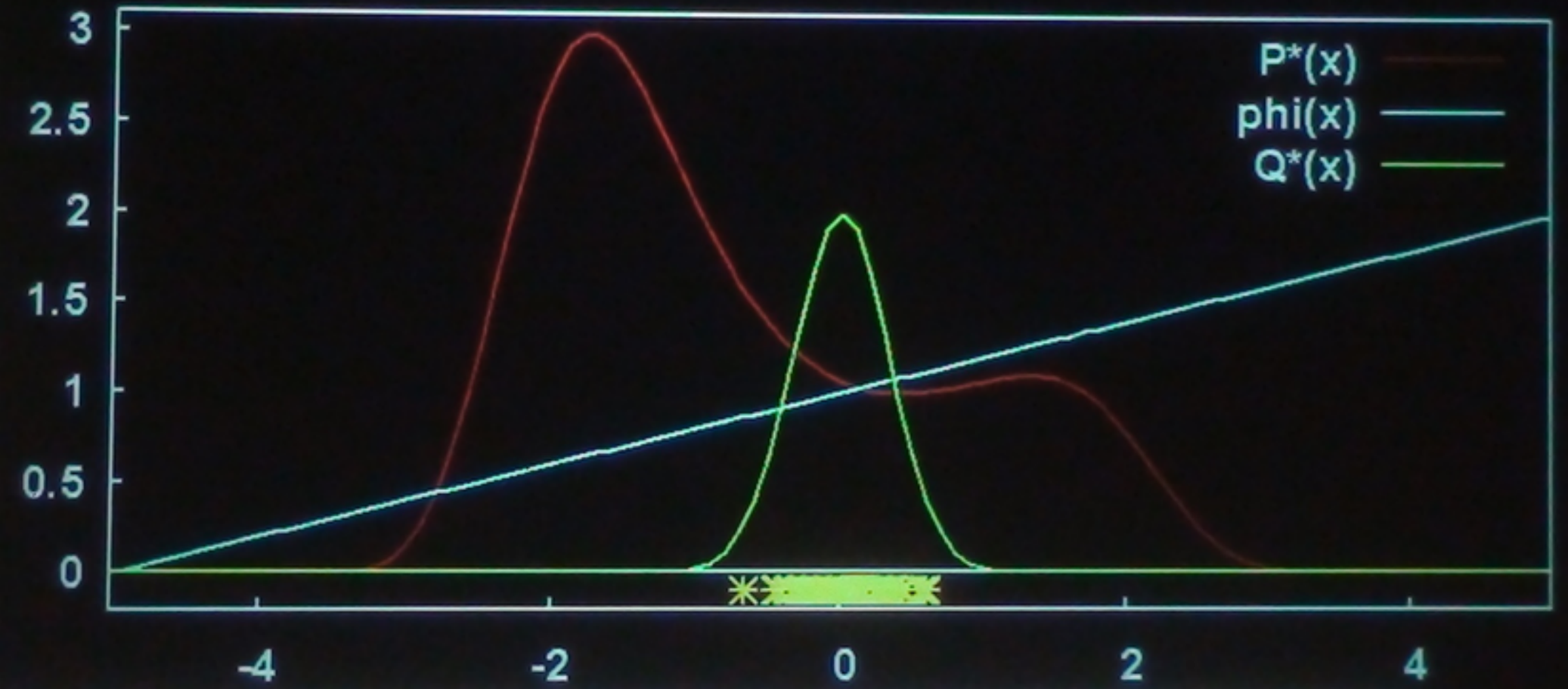
$$\frac{p^*(x^{(r)})}{Q(x^{(r)})}\,\phi(x^{(r)}) = \int \frac{p^*(x)}{Q(x)}\,\phi(x)\,Q(x)\,dx$$

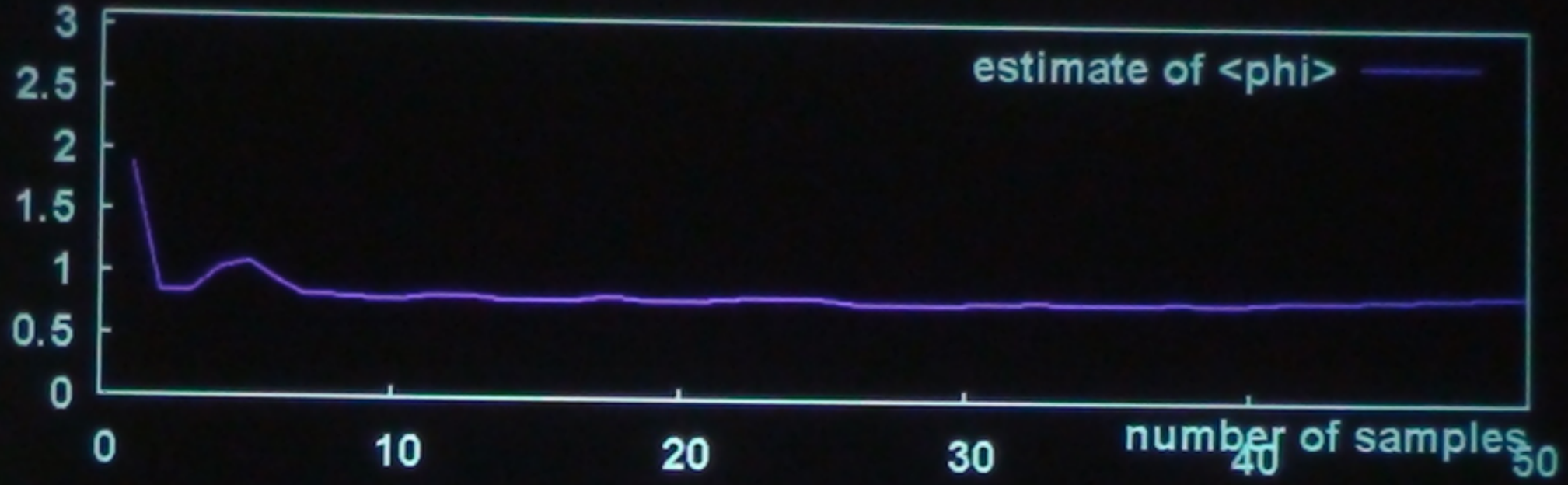
$$= \int dx\; p^*(x)\,\phi(x)$$

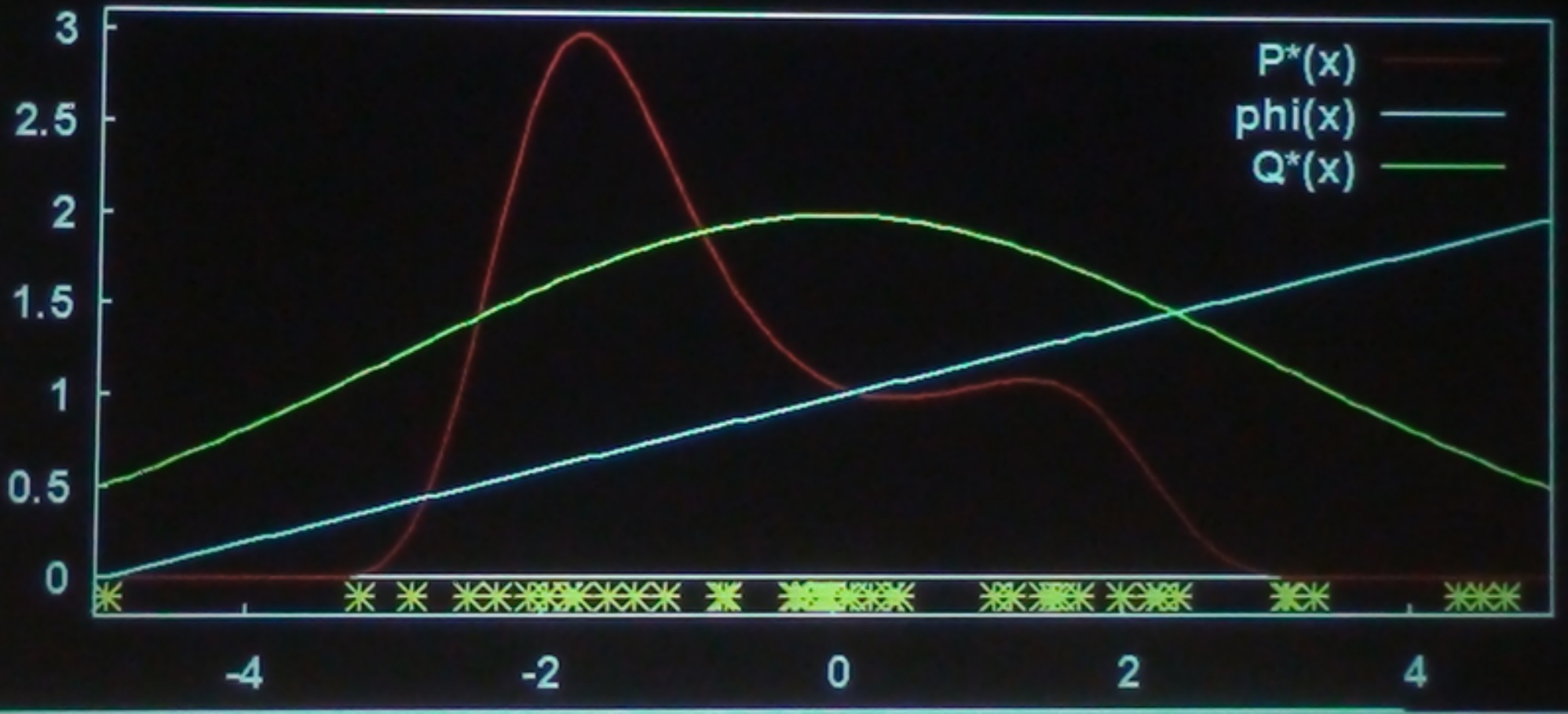

estimate of <phi>
3
2.5
2
1.5
1
0.5
0
0
10
20
30
40
50
number of samples

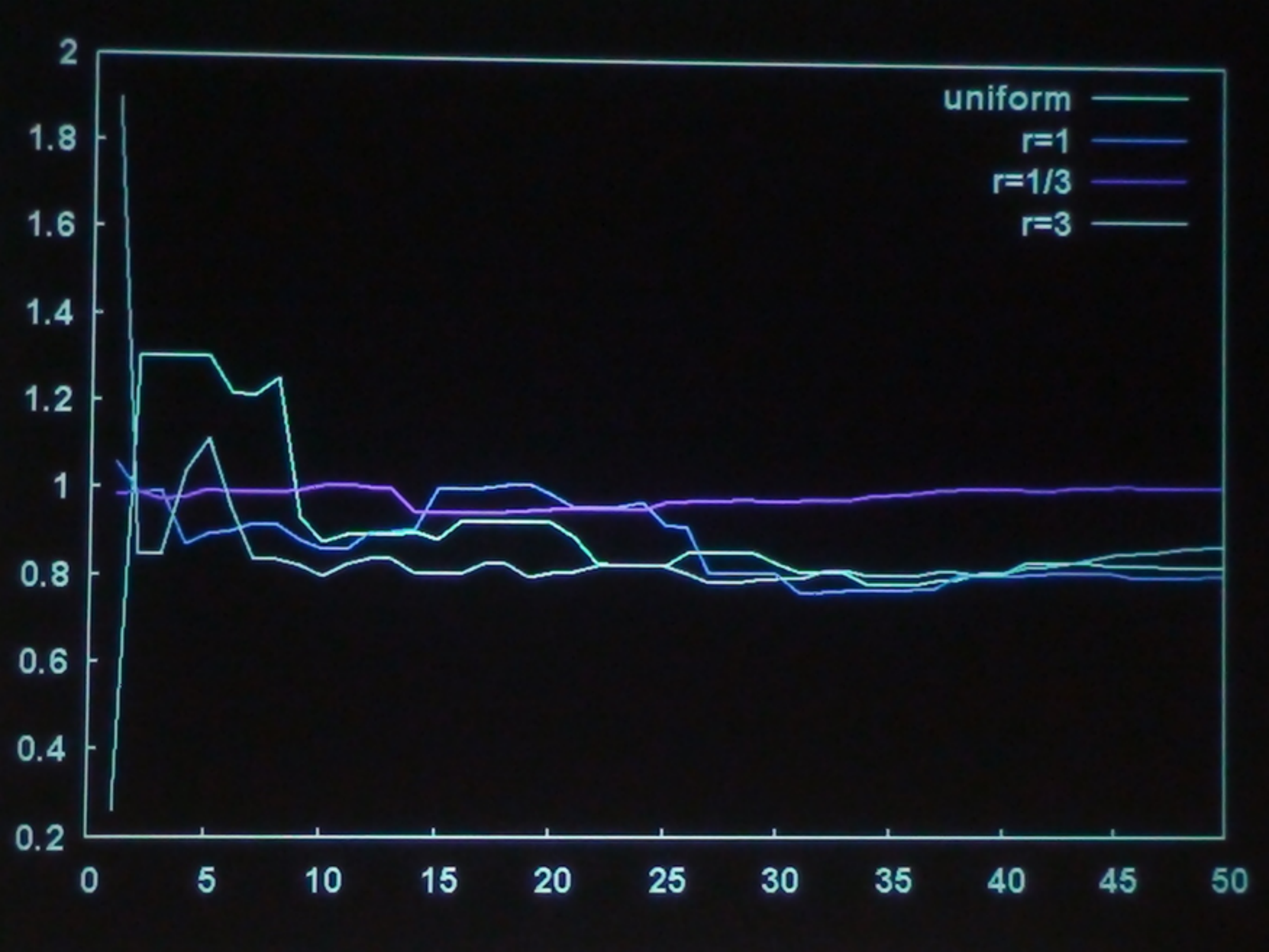

P*(x)
phi(x)
Q*(x)
3
2.5
2
1.5
1
0.5
0
-4
-2
0
2
4

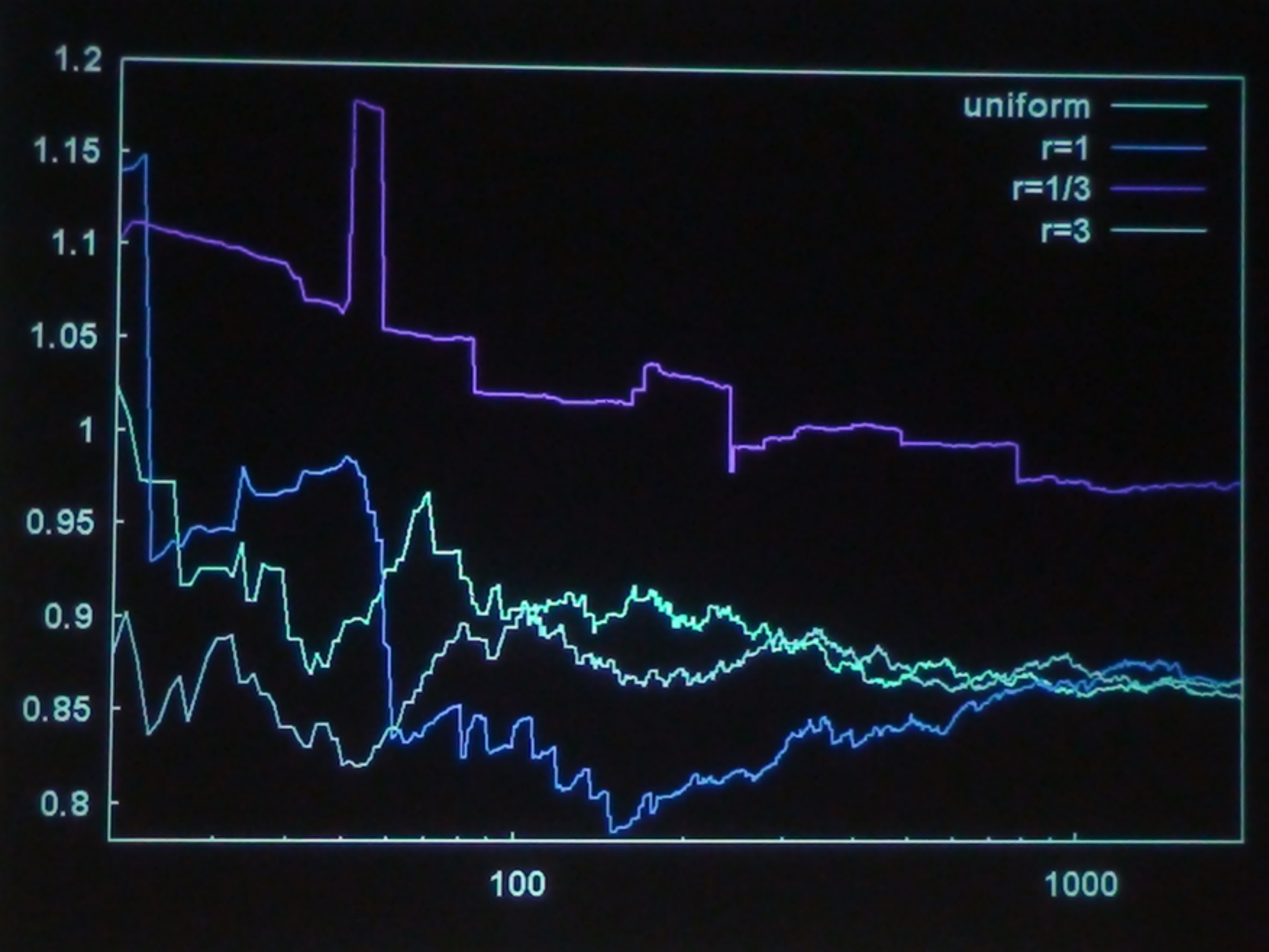

estimate of <phi>
3
2.5
2
1.5
1
0.5
0
0
10
20
30
40
50
number of samples

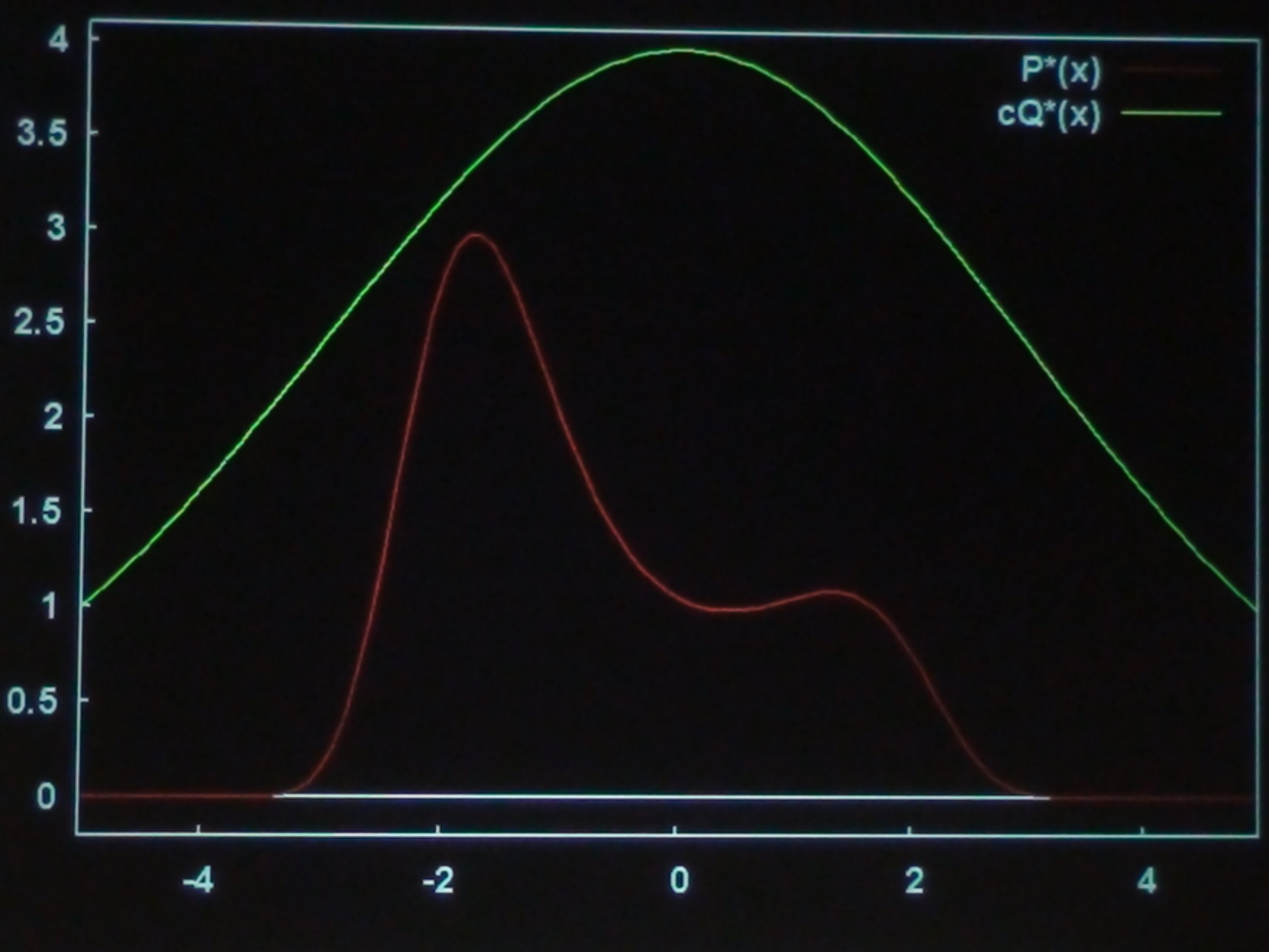

P*(x)
phi(x)
Q*(x)
3
2.5
2
1.5
1
0.5
0
-4
-2
0
2
4

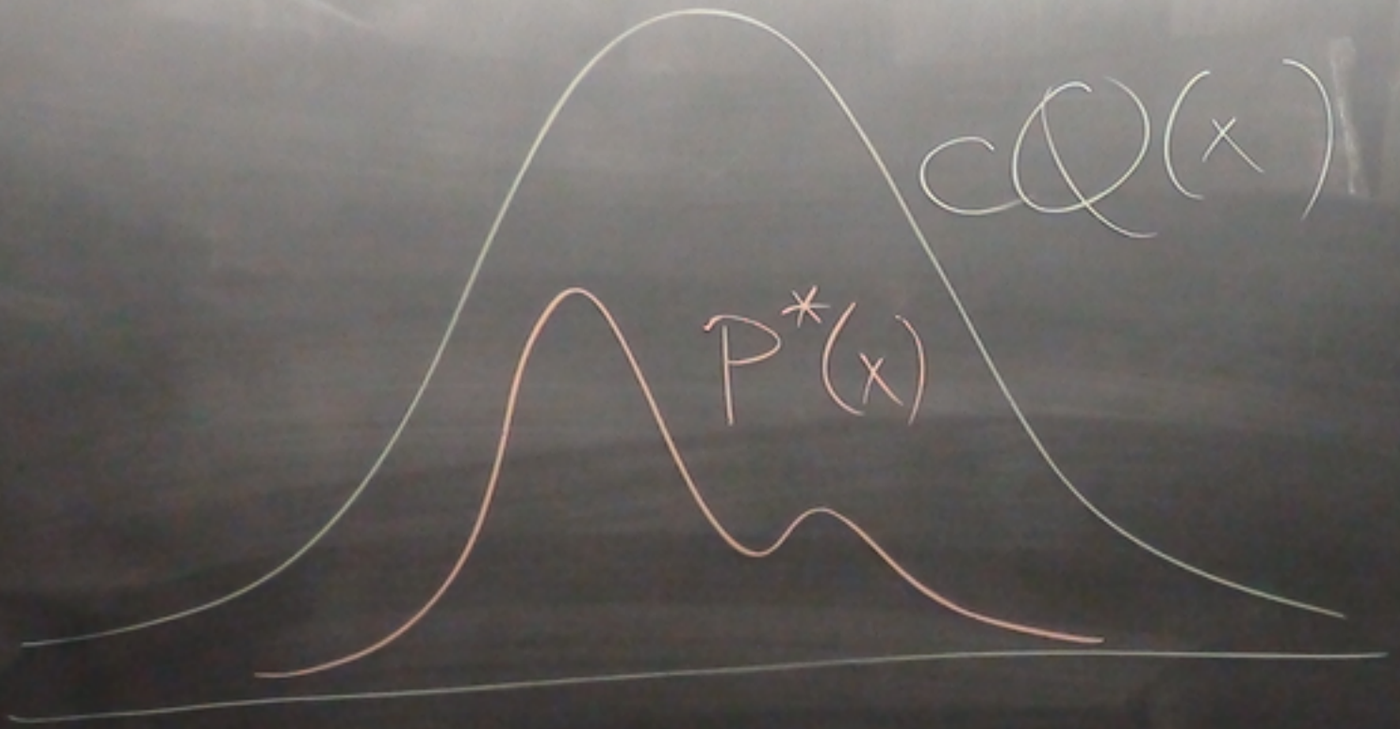

estimate of <phi>
3
2.5
2
1.5
1
0.5
0
0
10
20
30
40
50
number of samples


P*(x)
phi(x)
Q*(x)
3
2.5
2
1.5
1
0.5
0
-4
-2
0
2
4


uniform
r=1
r=1/3
r=3
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
5
10
15
20
25
30
35
40
45
50


1.2
1.15
1.1
1.05
1
0.95
0.9
0.85
0.8
100
1000
uniform
r=1
r=1/3
r=3


P*(x)
cQ*(x)
4
3.5
3
2.5
2
1.5
1
0.5
0
-4
-2
0
2
4



$cQ(x) \geq p^*(x)$

everywhere.

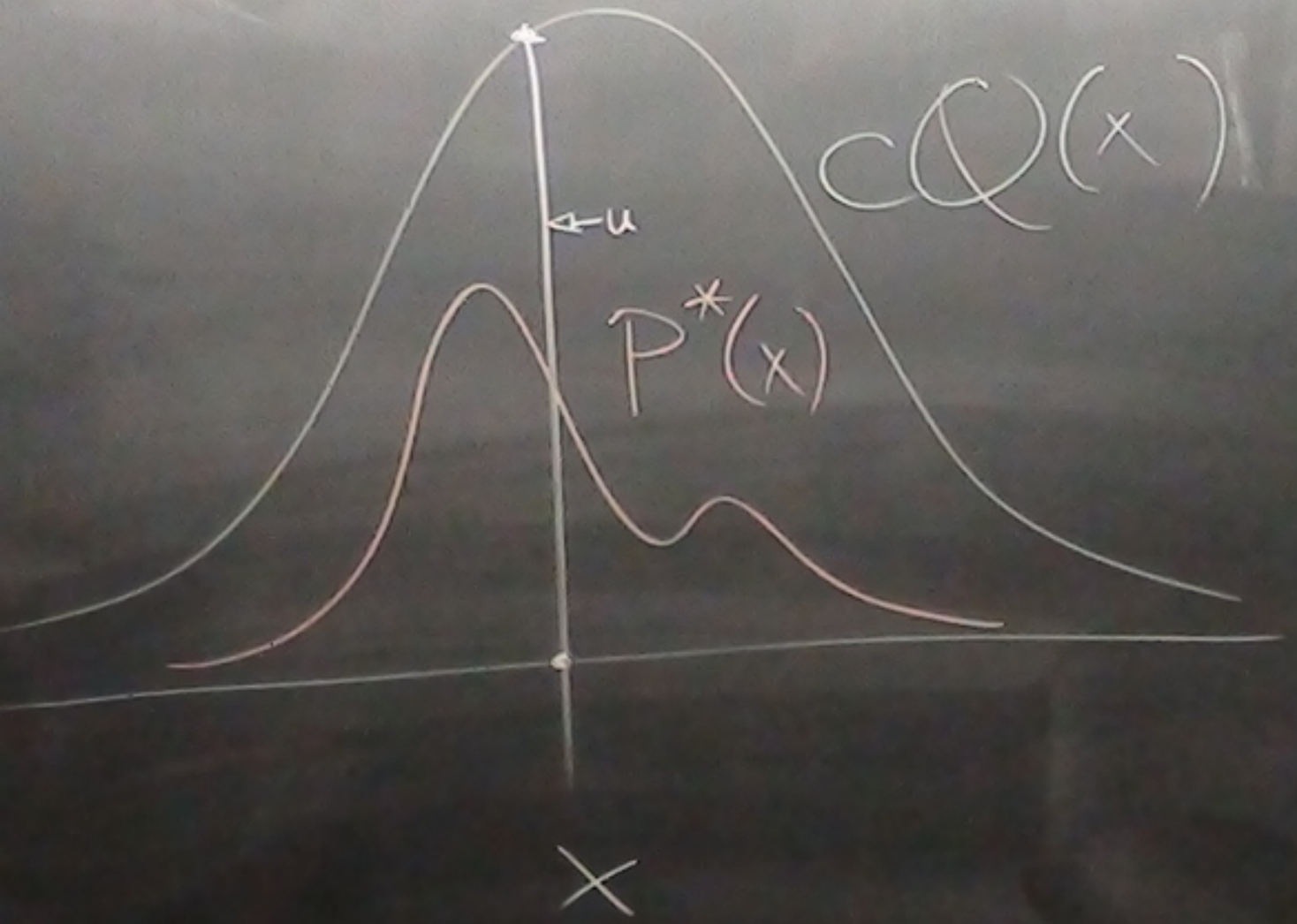
.)

everywhere.

1 $x \sim Q$

2 $u \sim \text{Uniform}(0, cQ(x))$

3 if $u < P^*(x)$ then "ACCEPT" x

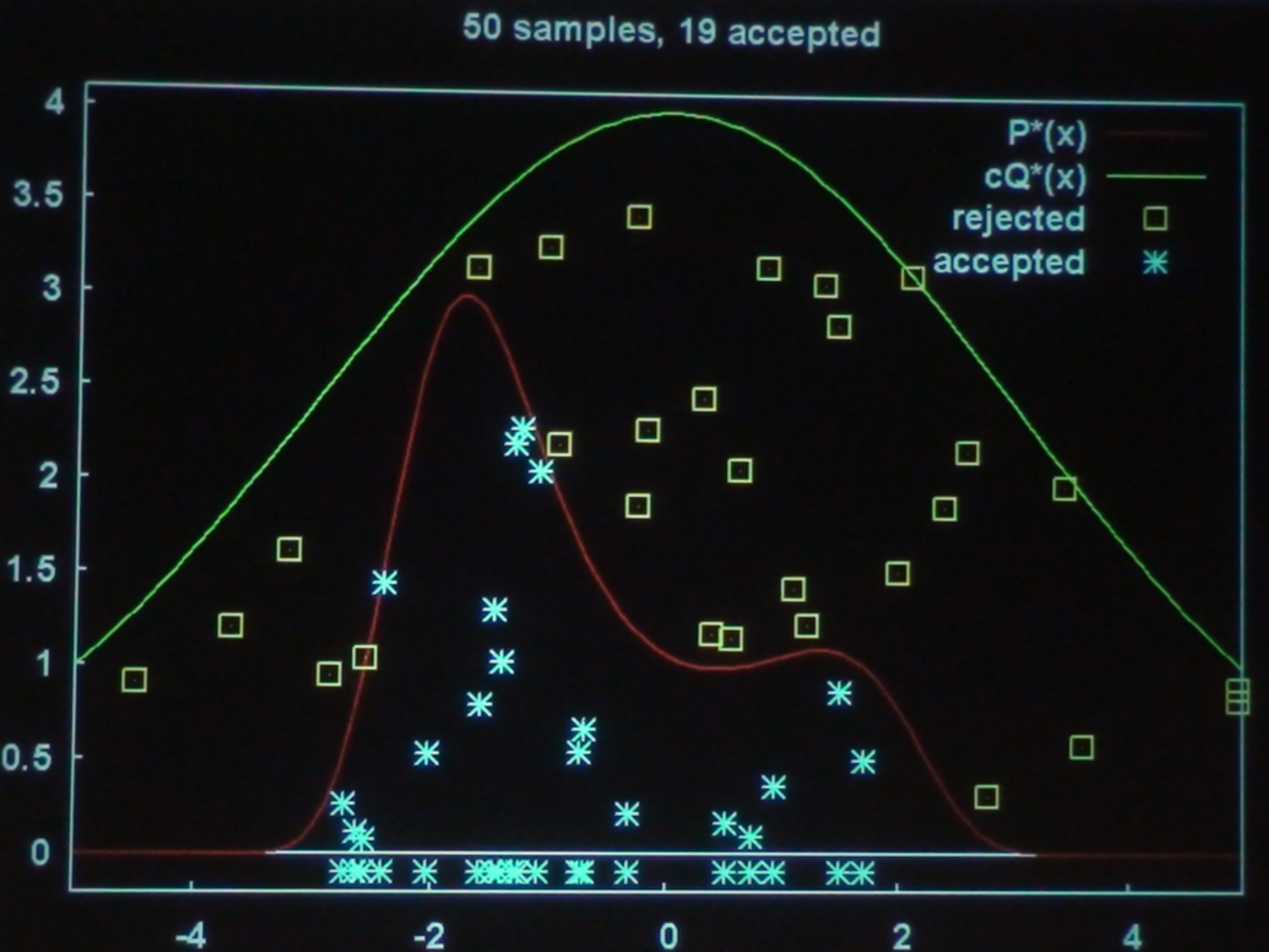


$cQ(x) \geq P^*(x)$ everywhere

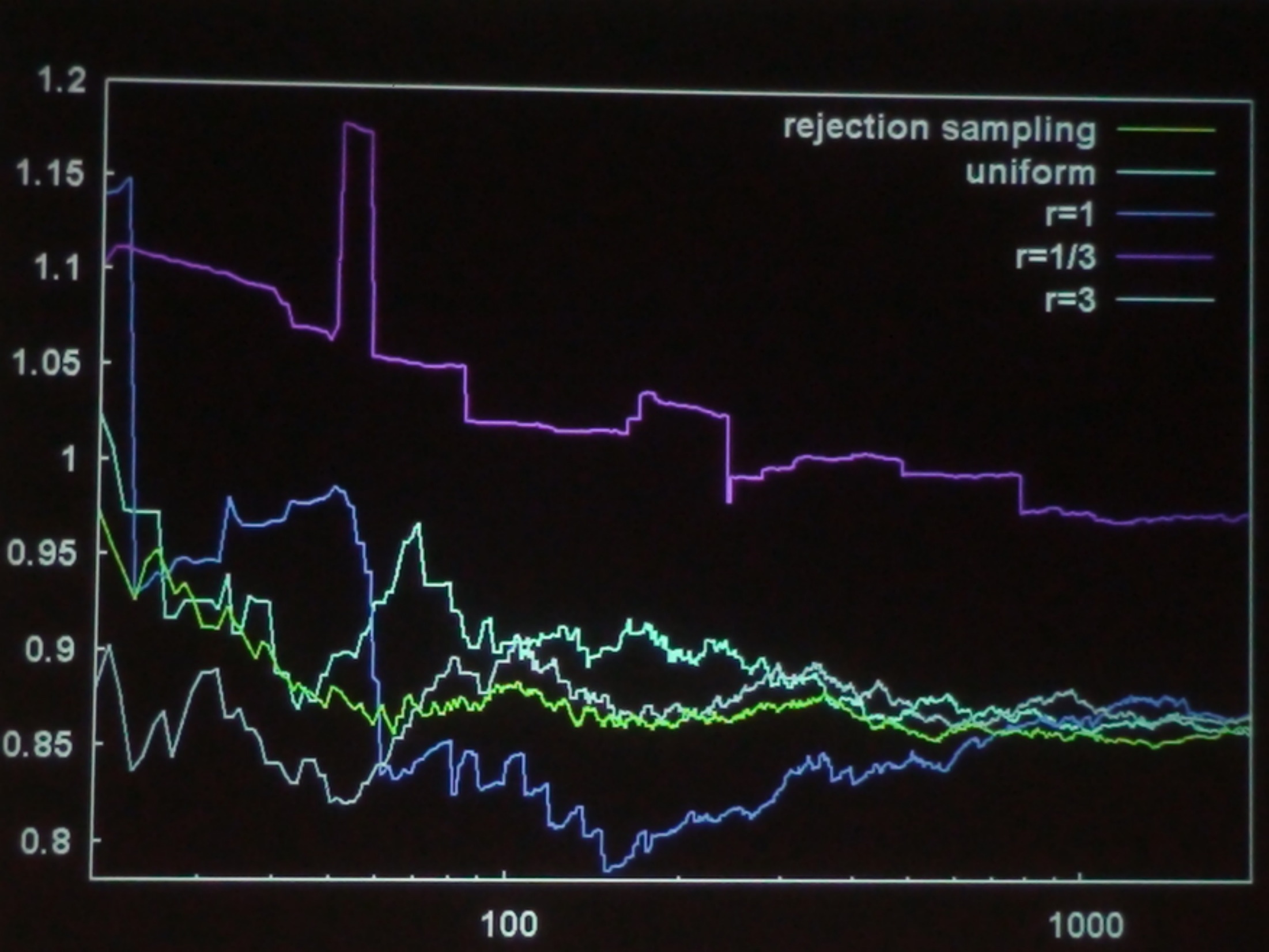
1

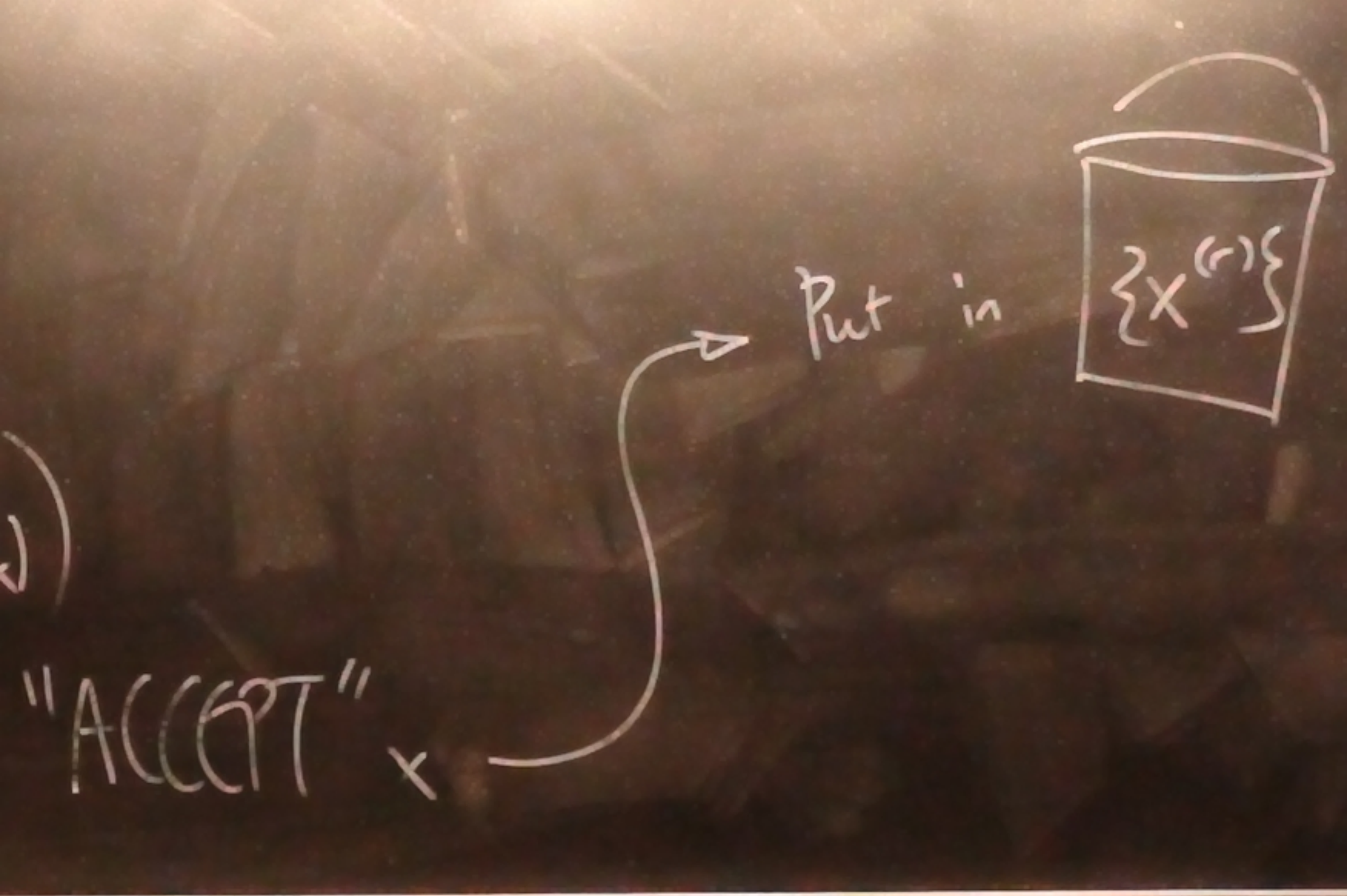
2

3


50 samples, 19 accepted
P*(x)
cQ*(x)
rejected
accepted
4
3.5
3
2.5
2
1.5
1
0.5
0
-4
-2
0
2
4

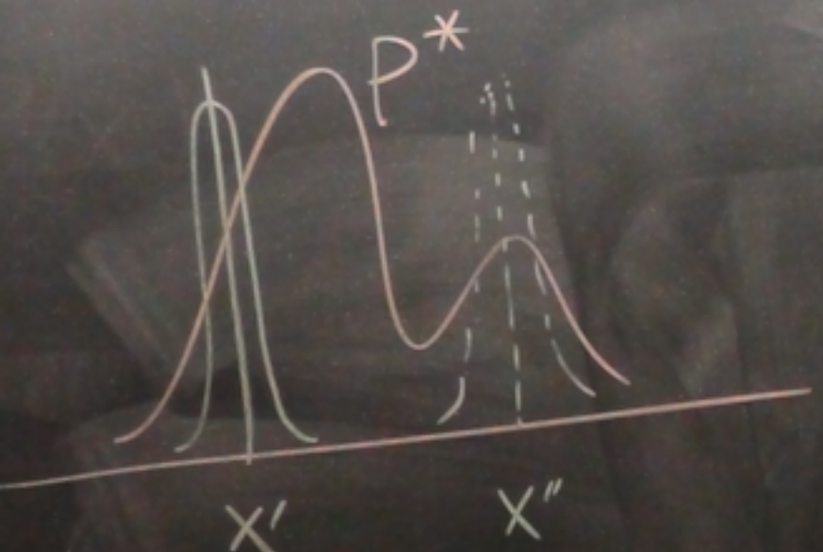
50 samples, 27 accepted
P*(x)
cQ*(x)
rejected
accepted
4
3.5
3
2.5
2
1.5
1
0.5
0
-4
-2
0
2
4


1.2
1.15
1.1
1.05
1
0.95
0.9
0.85
0.8
100
1000
rejection sampling
uniform
r=1
r=1/3
r=3


Put in
$\{x^{(r)}\}$
"ACCEPT" x

Markov Chain Monte Carlo

Metropolis



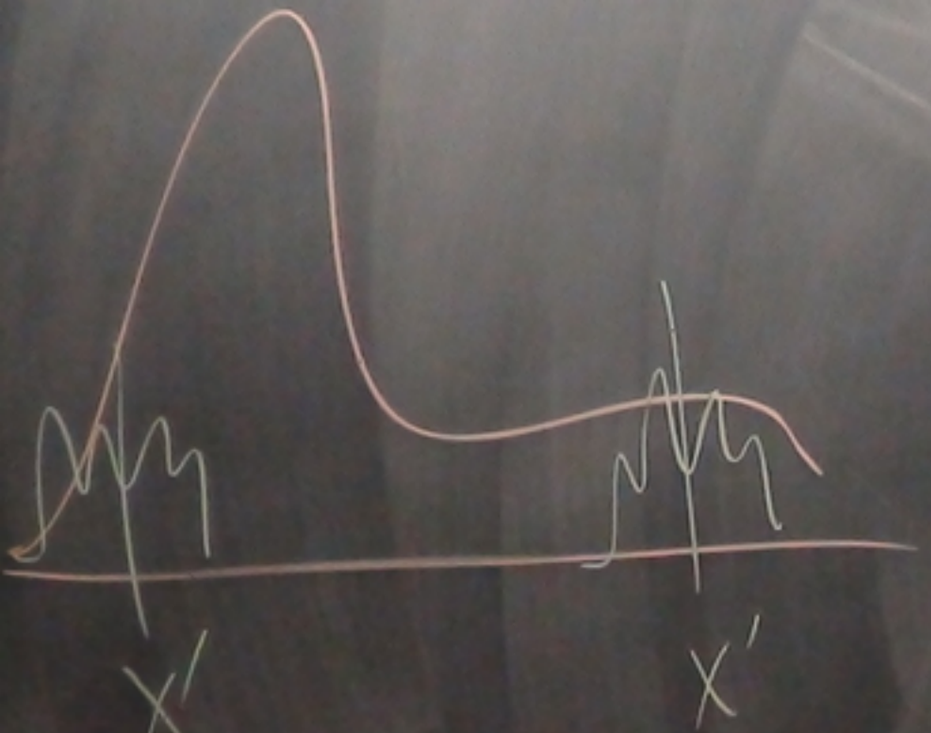
$Q(X; X')$

next location

current location

OK for Q to be simple or complicated

OK for Q to not look like P



if it's going to work
then usually Q makes
a small change to $(-\log P^*(x))$

At time t, at $X^{(t)}$,

1. generate $x \sim Q(x; x^{(t)})$

2. compute $a = \dfrac{P^*(x)\, Q(x^{(t)}; x)}{P^*(x^{(t)})\, Q(x; x^{(t)})}$

"acceptance ratio"

t, at $x^{(t)}$,

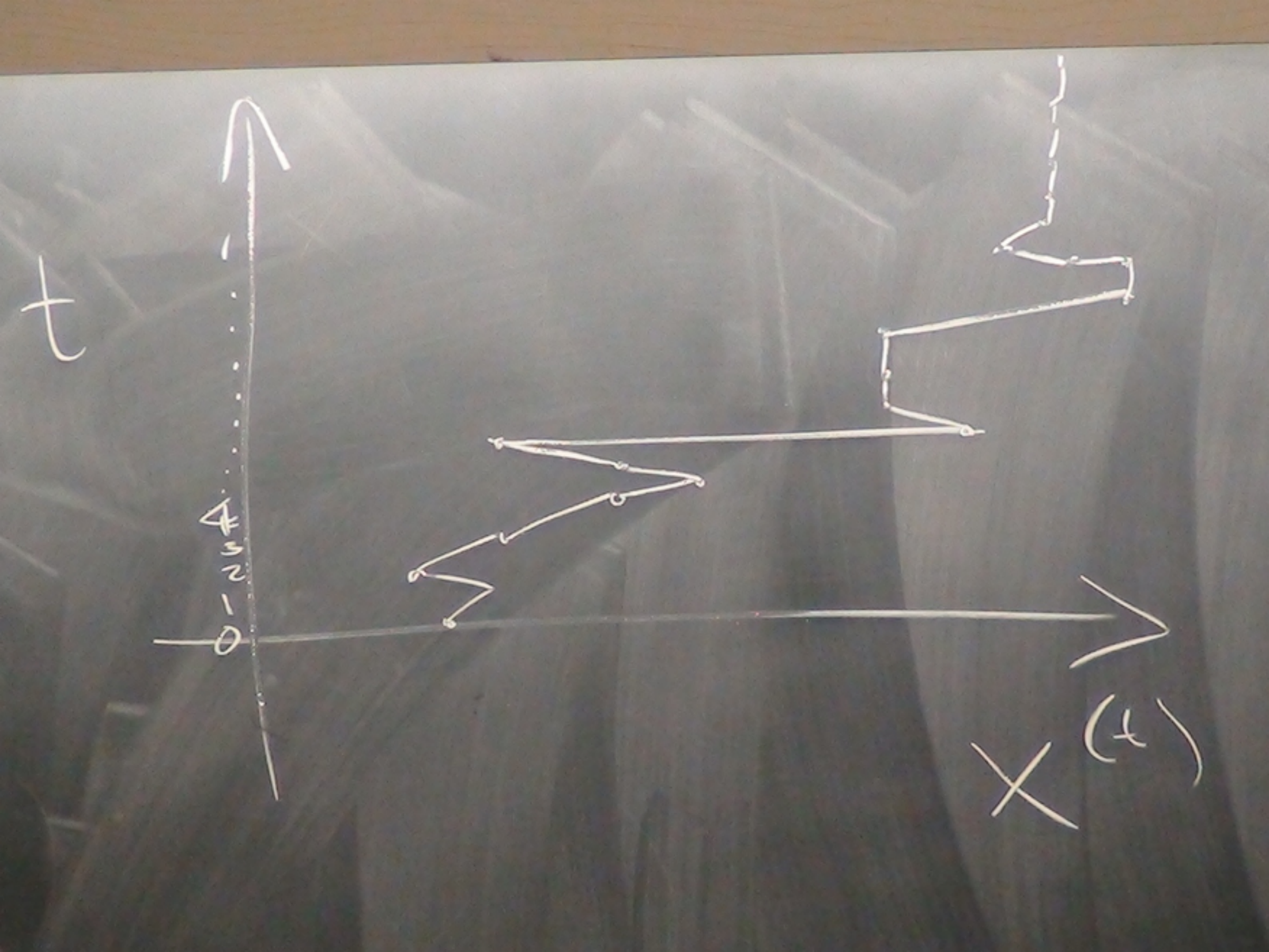
$Q = \frac{Q^*}{Z(x^{(t)})}$

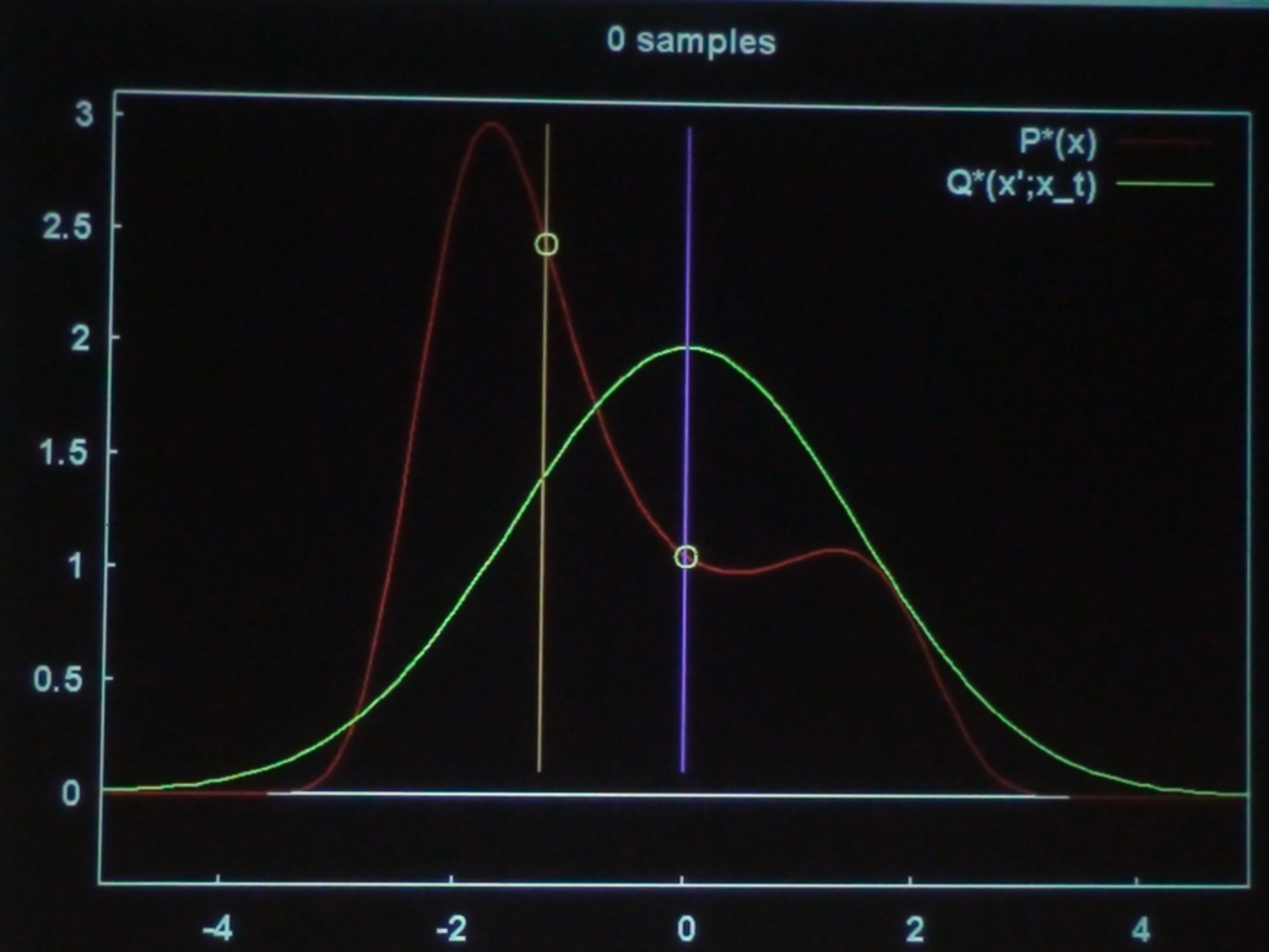
1. generate $x \sim Q(x; x^{(t)})$

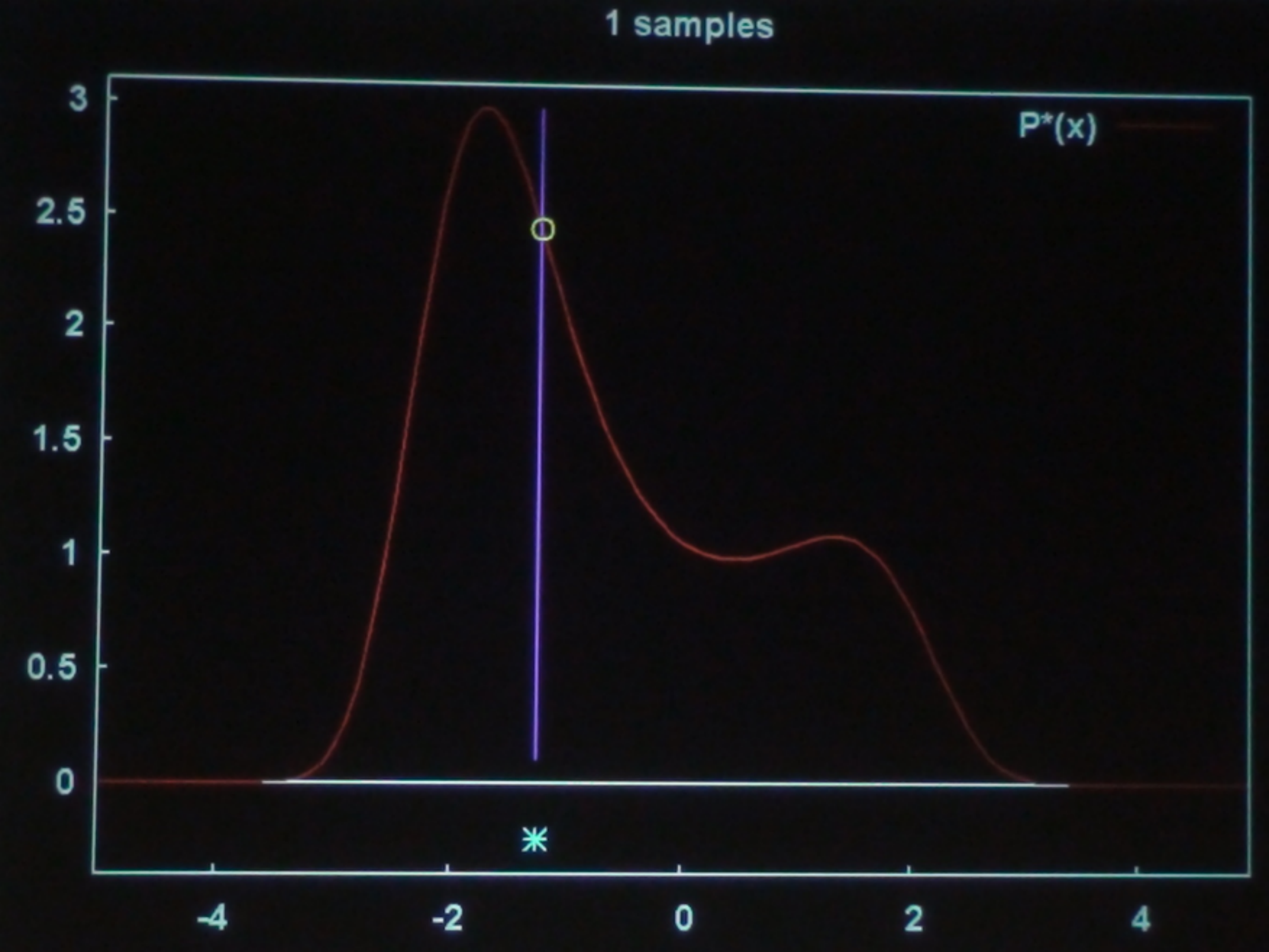
2. compute $a = \frac{P^*(x)\, Q(x^{(t)}; x)}{P^*(x^{(t)})\, Q(x; x^{(t)})}$

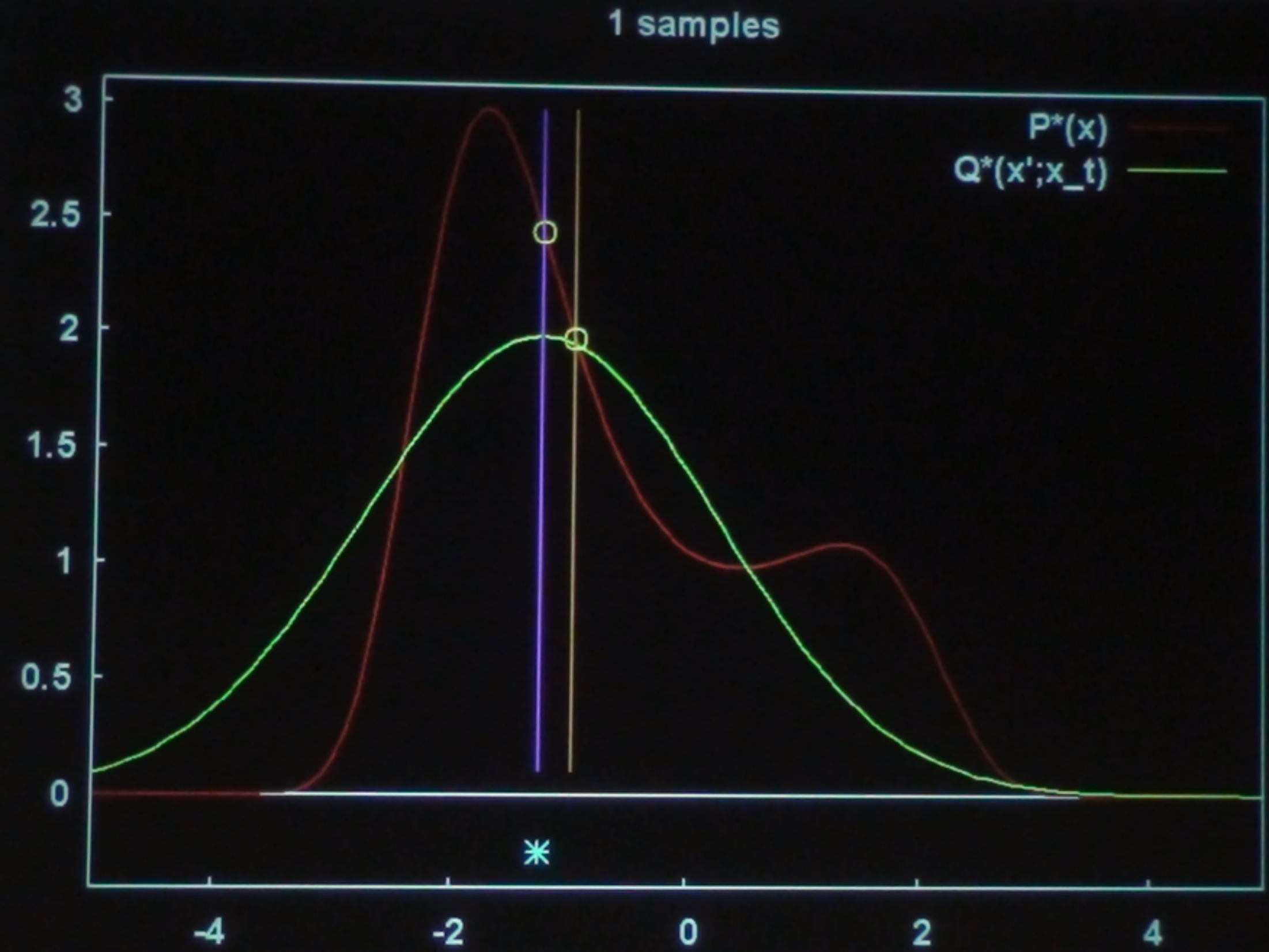
"acceptance ratio"

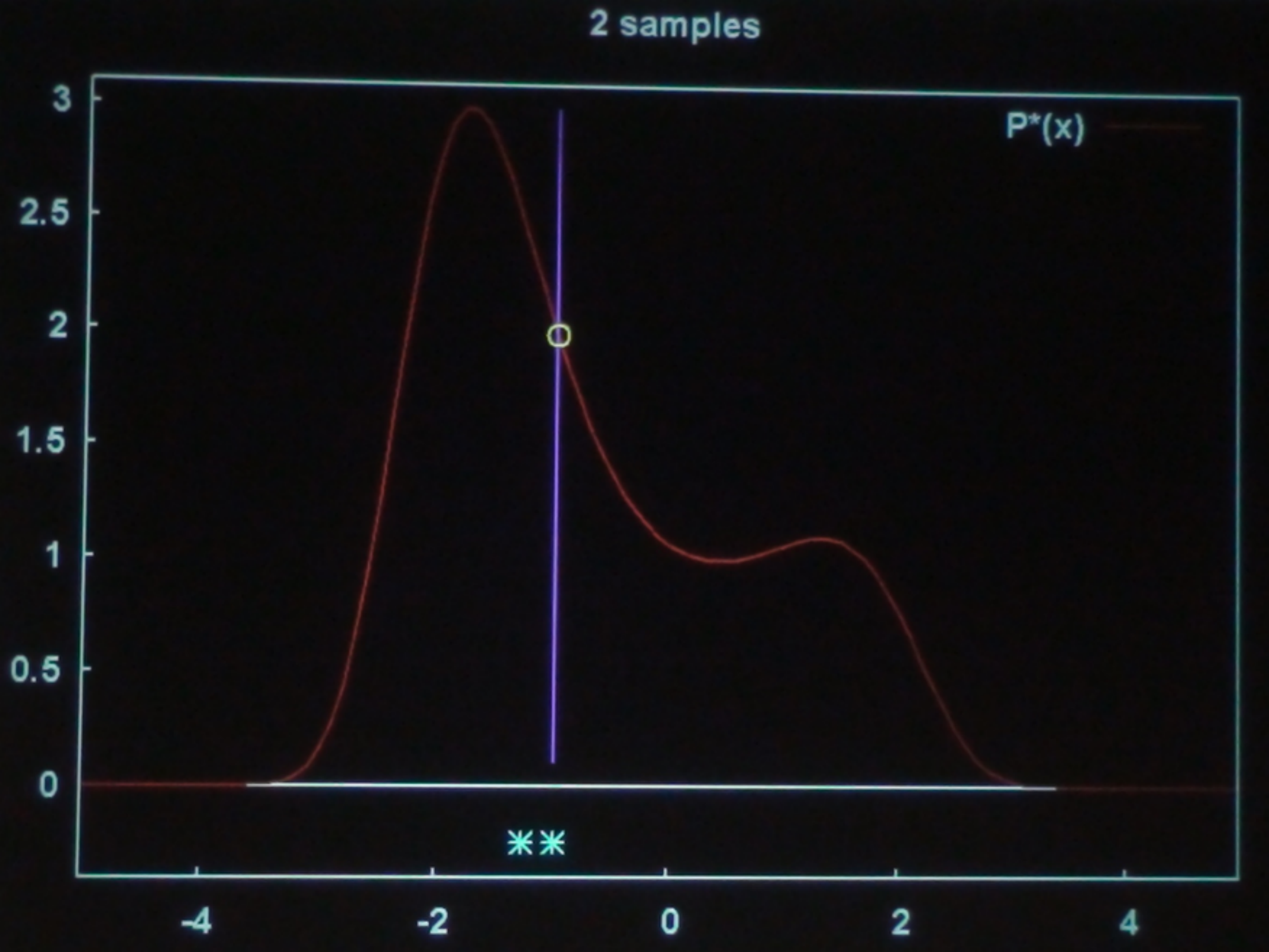
3. if $a > 1$ accept x accept: $x^{(t+1)} = x$
otherwise accept x with probability a. reject: $x^{(t+1)} = x^{(t)}$

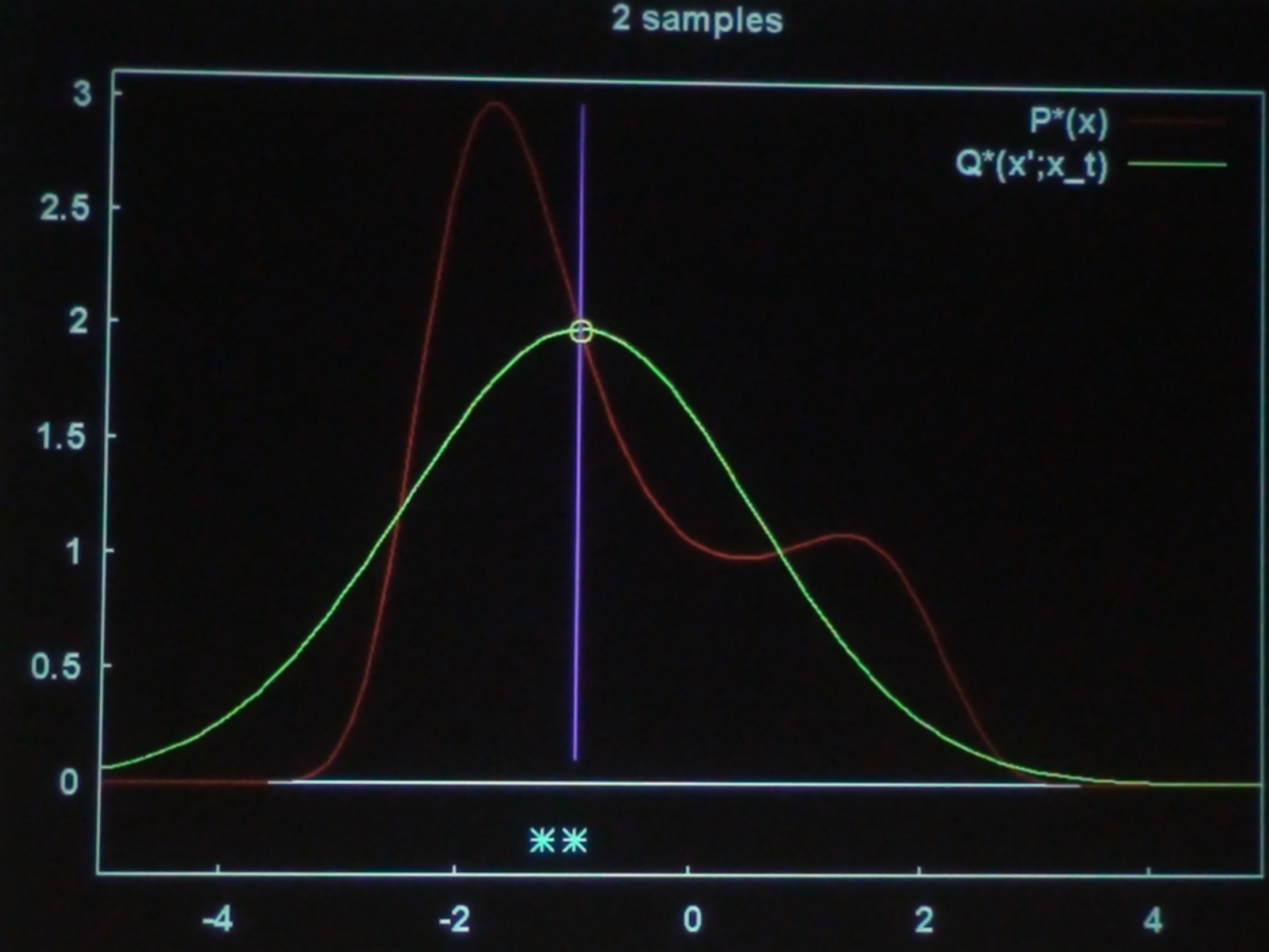

t
0
1
2
3
4
X(t)

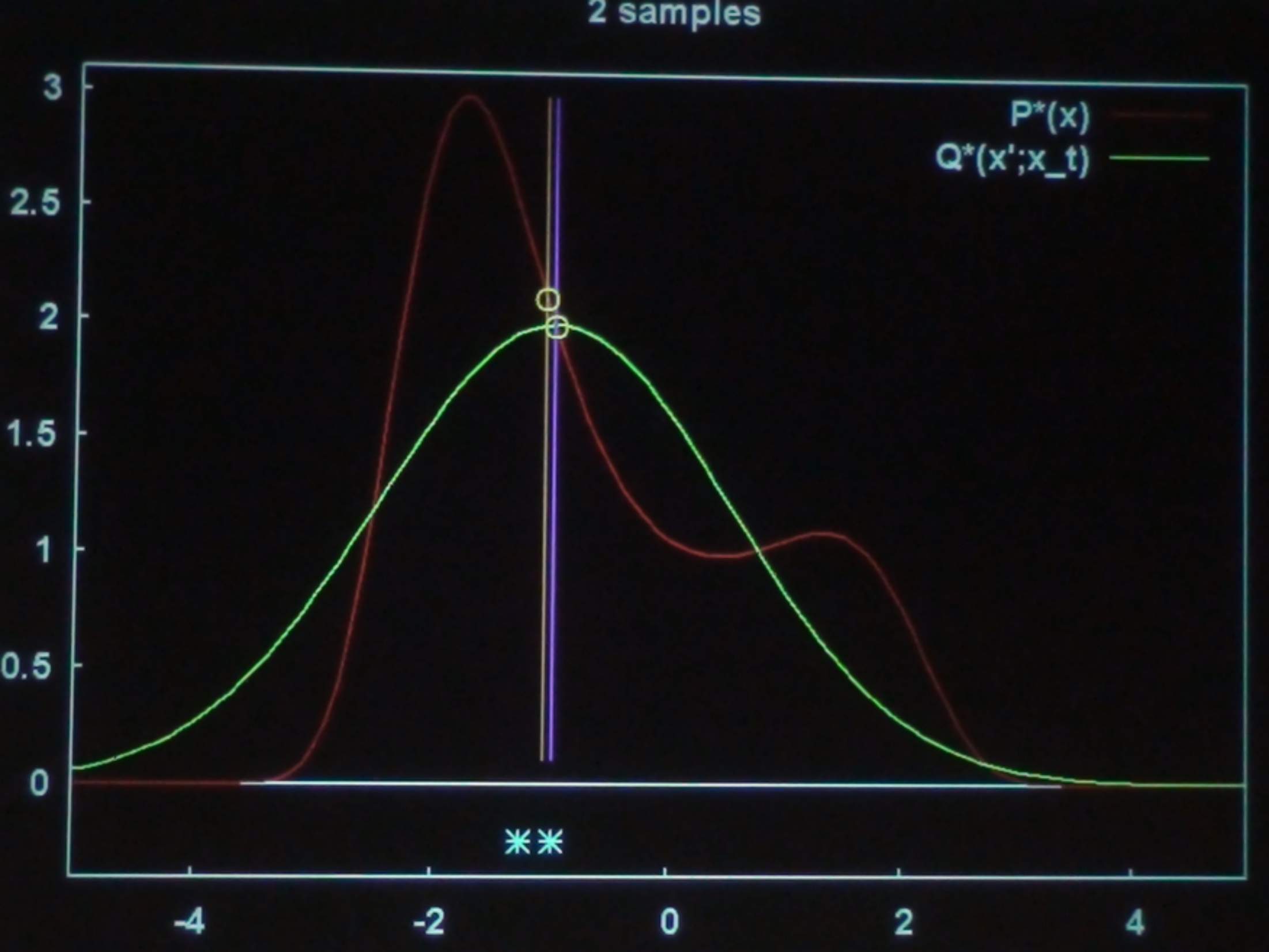

0 samples
P*(x)
Q*(x';x_t)
3
2.5
2
1.5
1
0.5
0
-4
-2
0
2
4

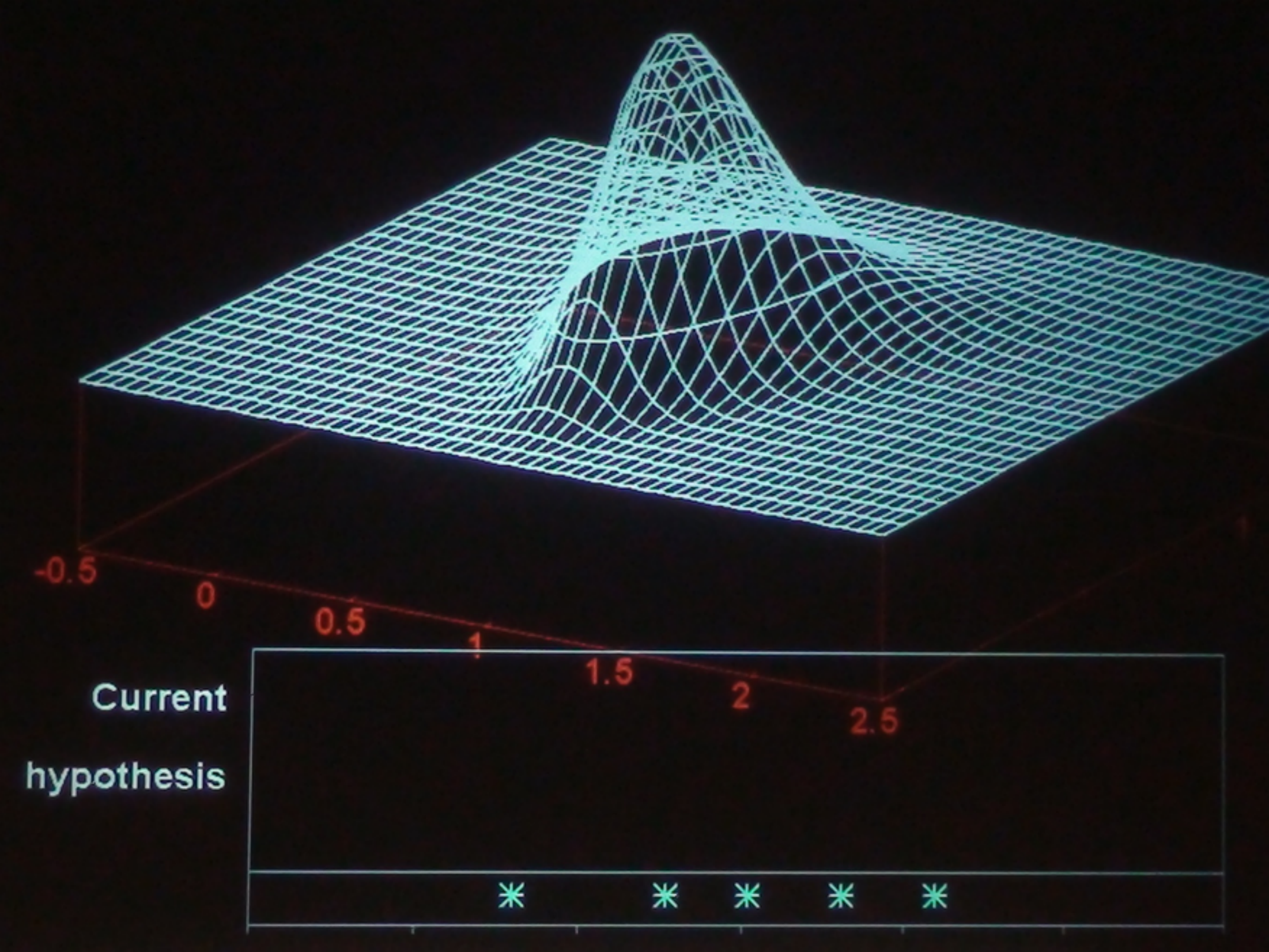

1 samples
P*(x)
3
2.5
2
1.5
1
0.5
0
-4
-2
0
2
4

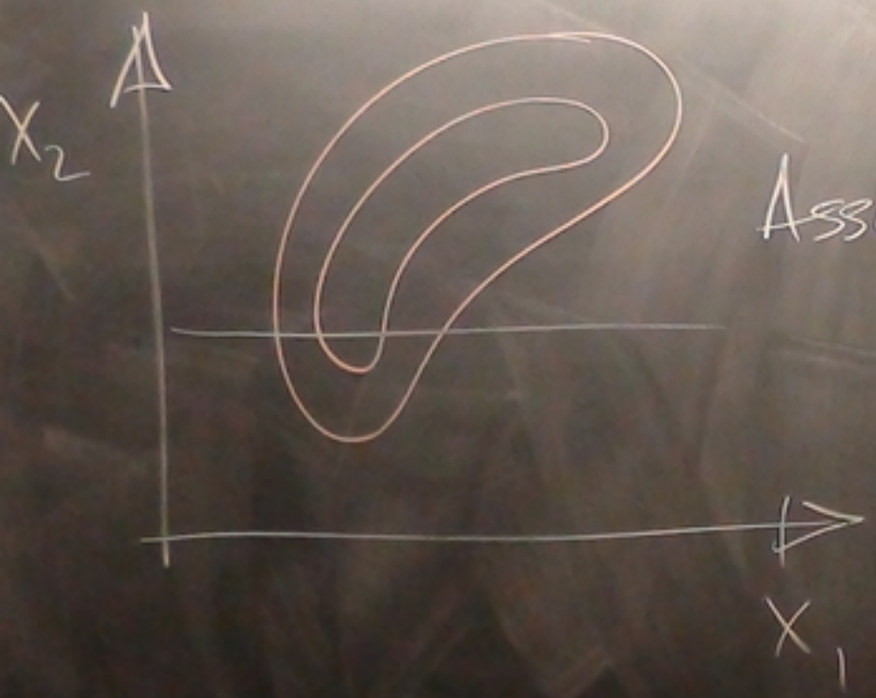

1 samples
P*(x)
Q*(x';x_t)
3
2.5
2
1.5
1
0.5
0
-4
-2
0
2
4


2 samples
P*(x)
3
2.5
2
1.5
1
0.5
0
-4
-2
0
2
4


2 samples
P*(x)
Q*(x';x_t)
3
2.5
2
1.5
1
0.5
0
-4
-2
0
2
4


2 samples
P*(x)
Q*(x';x_t)
3
2.5
2
1.5
1
0.5
0
-4
-2
0
2
4


-0.5
0
0.5
1
1.5
2
2.5
Current
hypothesis



Assume that

can sample from

conditional distributions

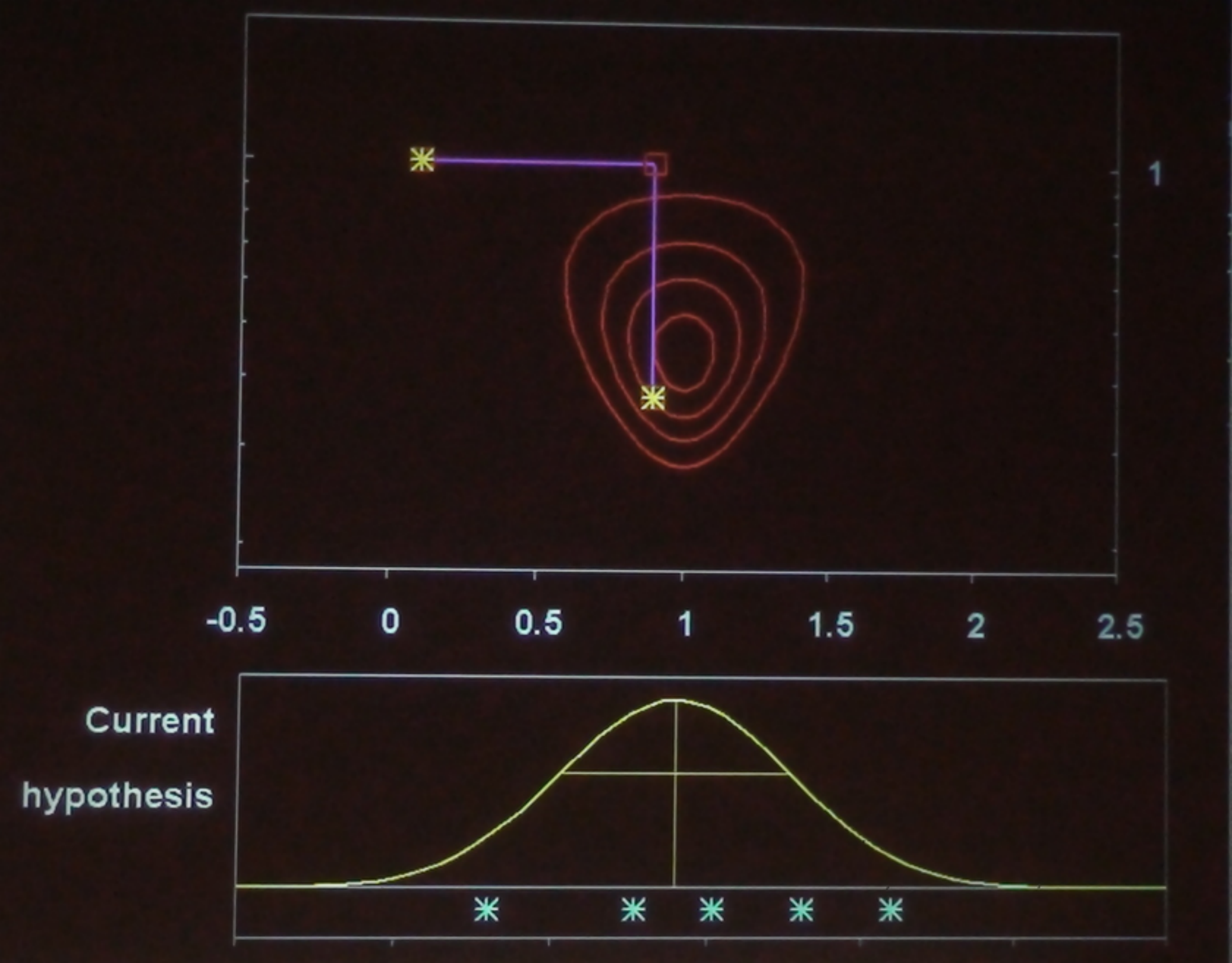
$P(x_1 | x_2)$

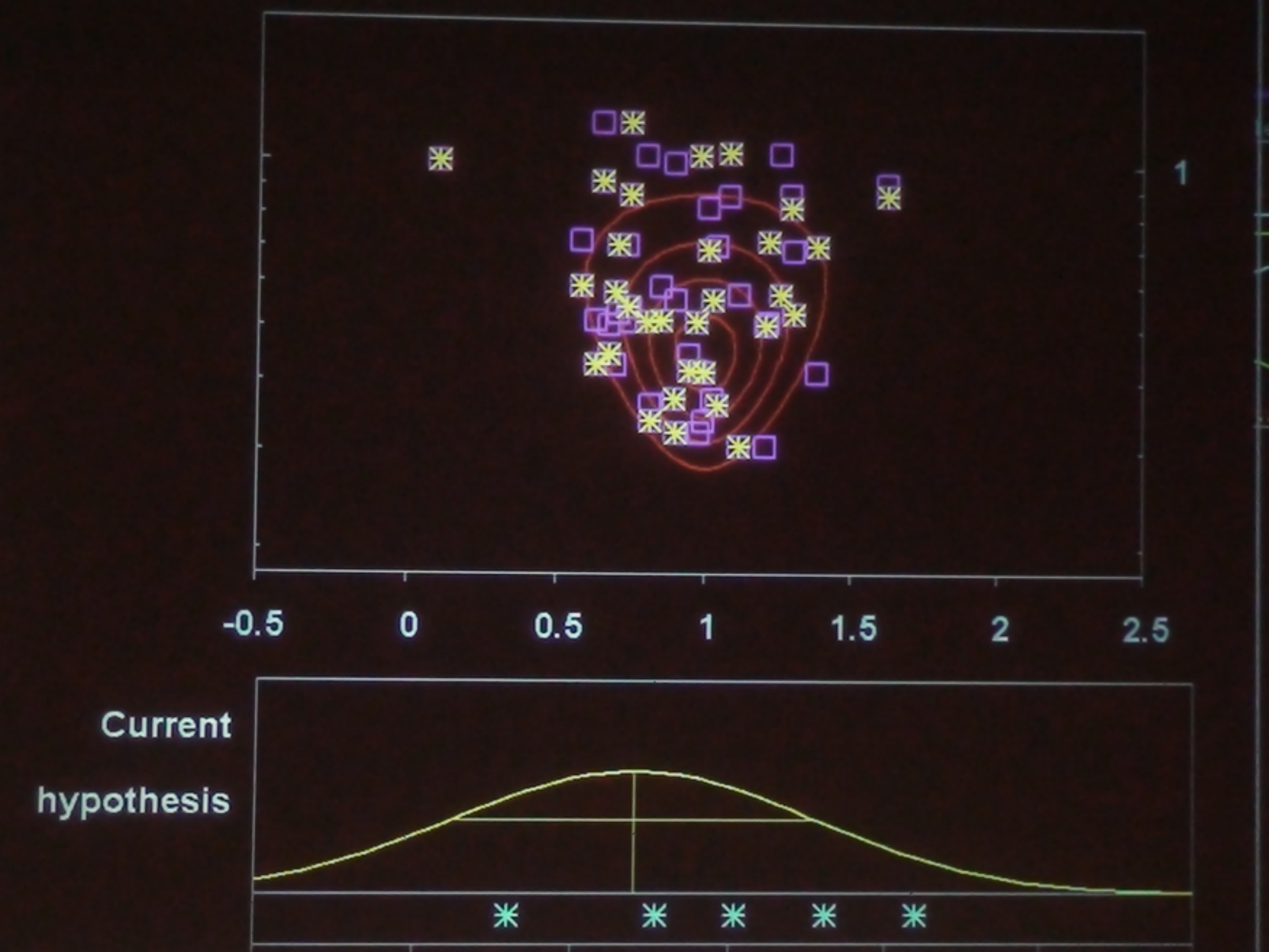
& $P(x_2 | x_1)$

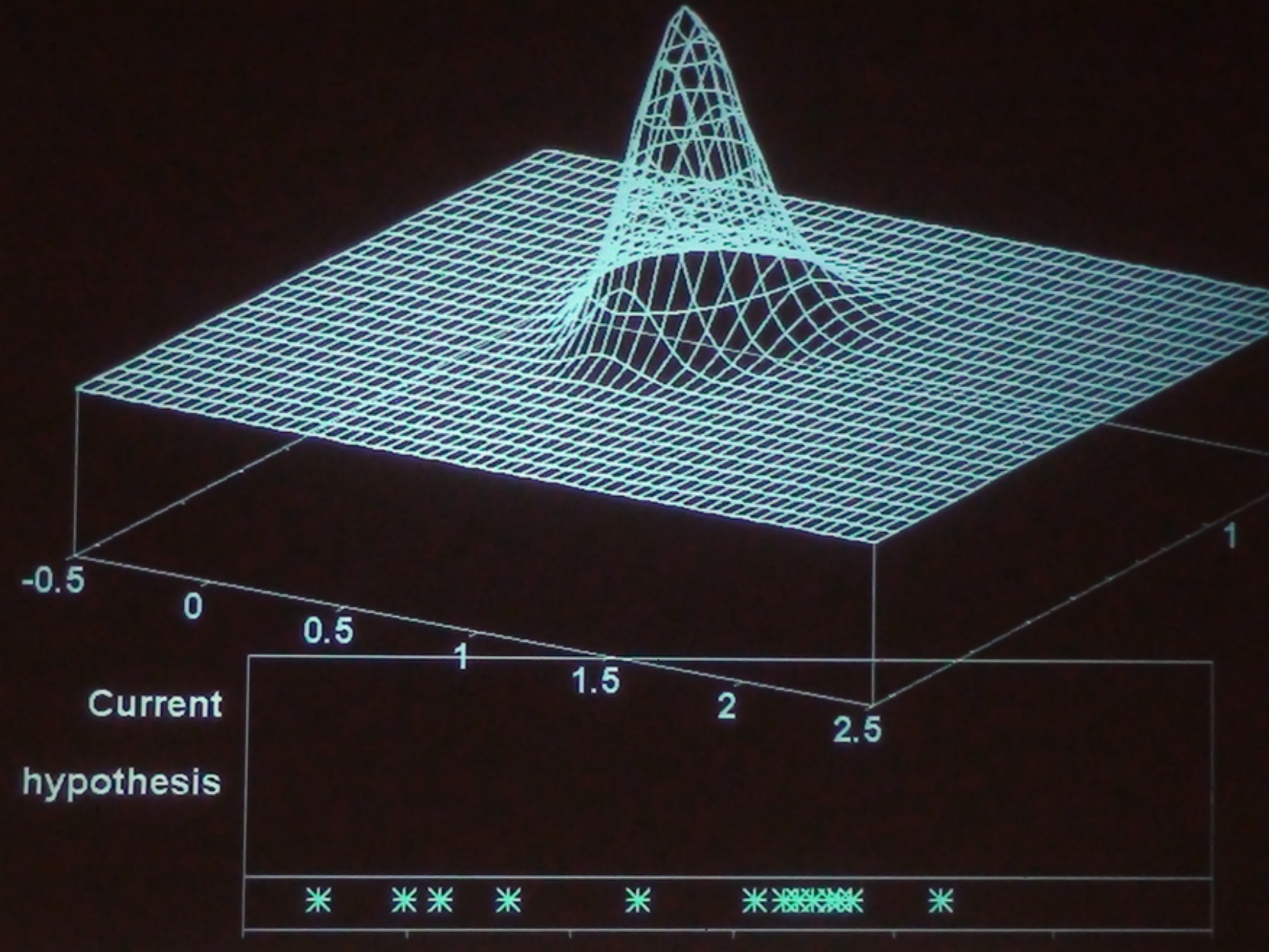
start at $X^{(t)}$

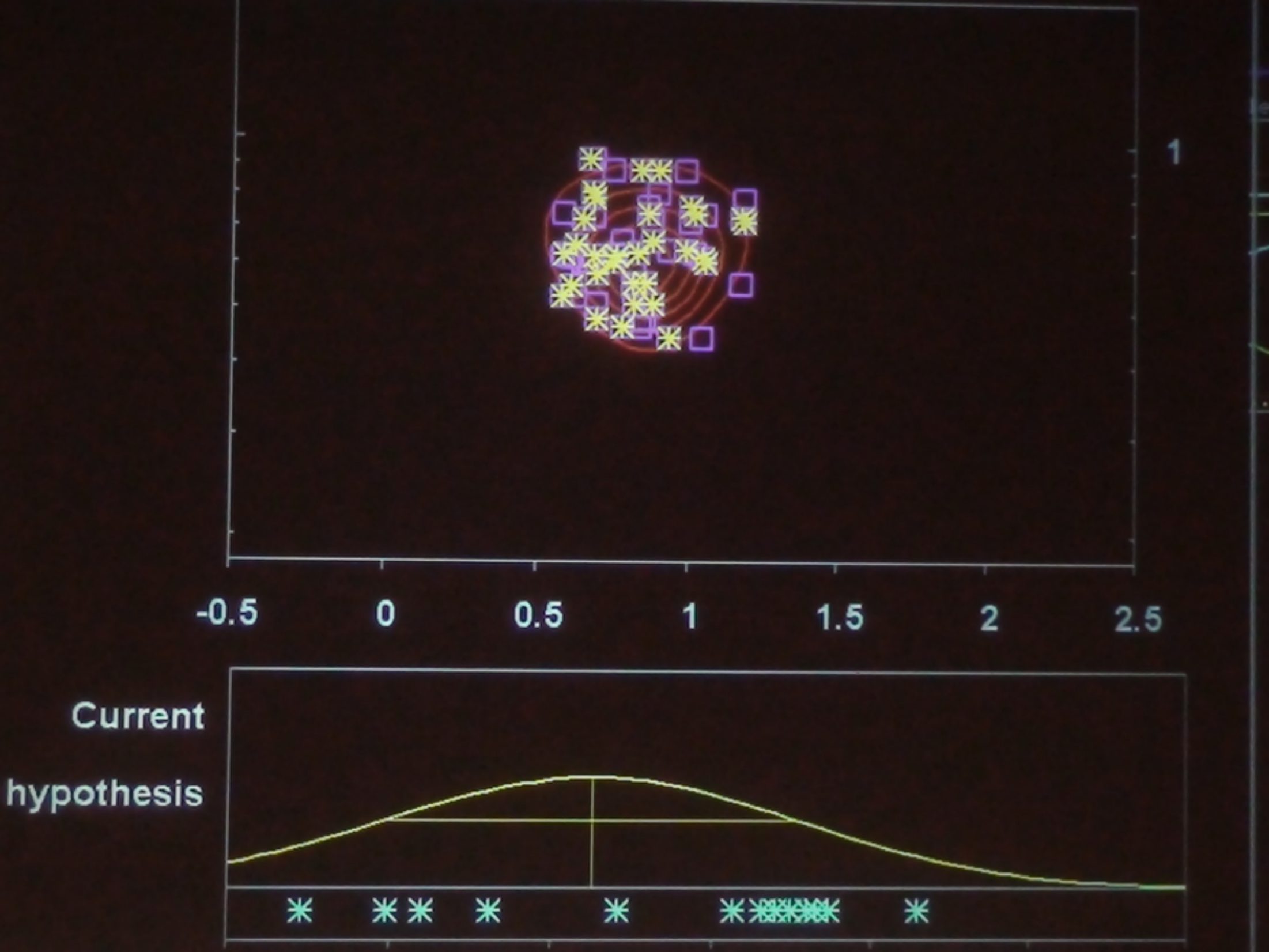
$$X_1^{t+1} \sim P(x_1 \mid X_2^t)$$

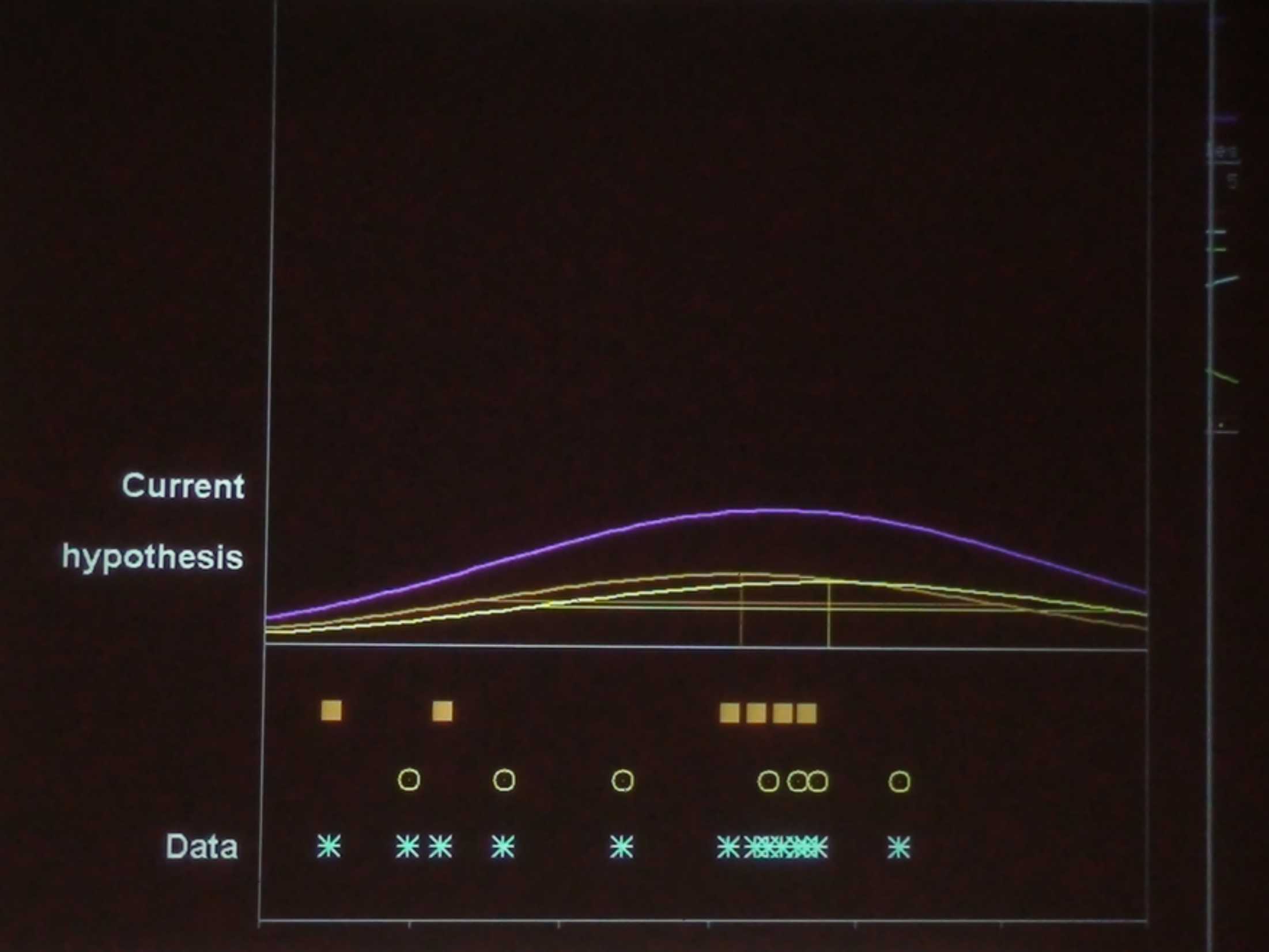
$$X_2^{t+1} \sim P(x_2 \mid x_1^{t+1})$$


1
-0.5
0
0.5
1
1.5
2
2.5
Current
hypothesis


1
-0.5
0
0.5
1
1.5
2
2.5
Current
hypothesis


-0.5
0
0.5
1
1.5
2
2.5
1
Current
hypothesis


1
-0.5
0
0.5
1
1.5
2
2.5
Current
hypothesis

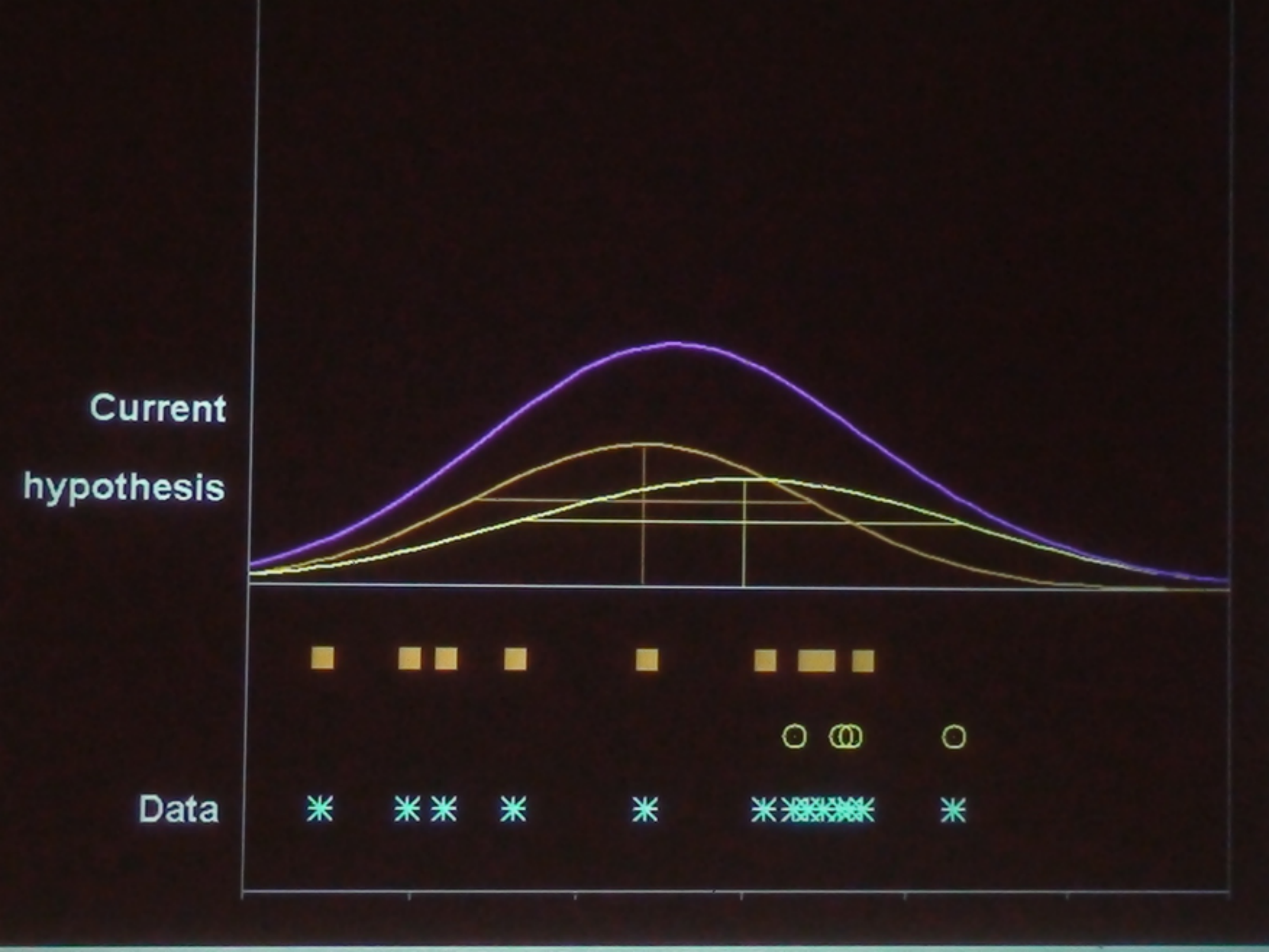

Current
hypothesis
Data

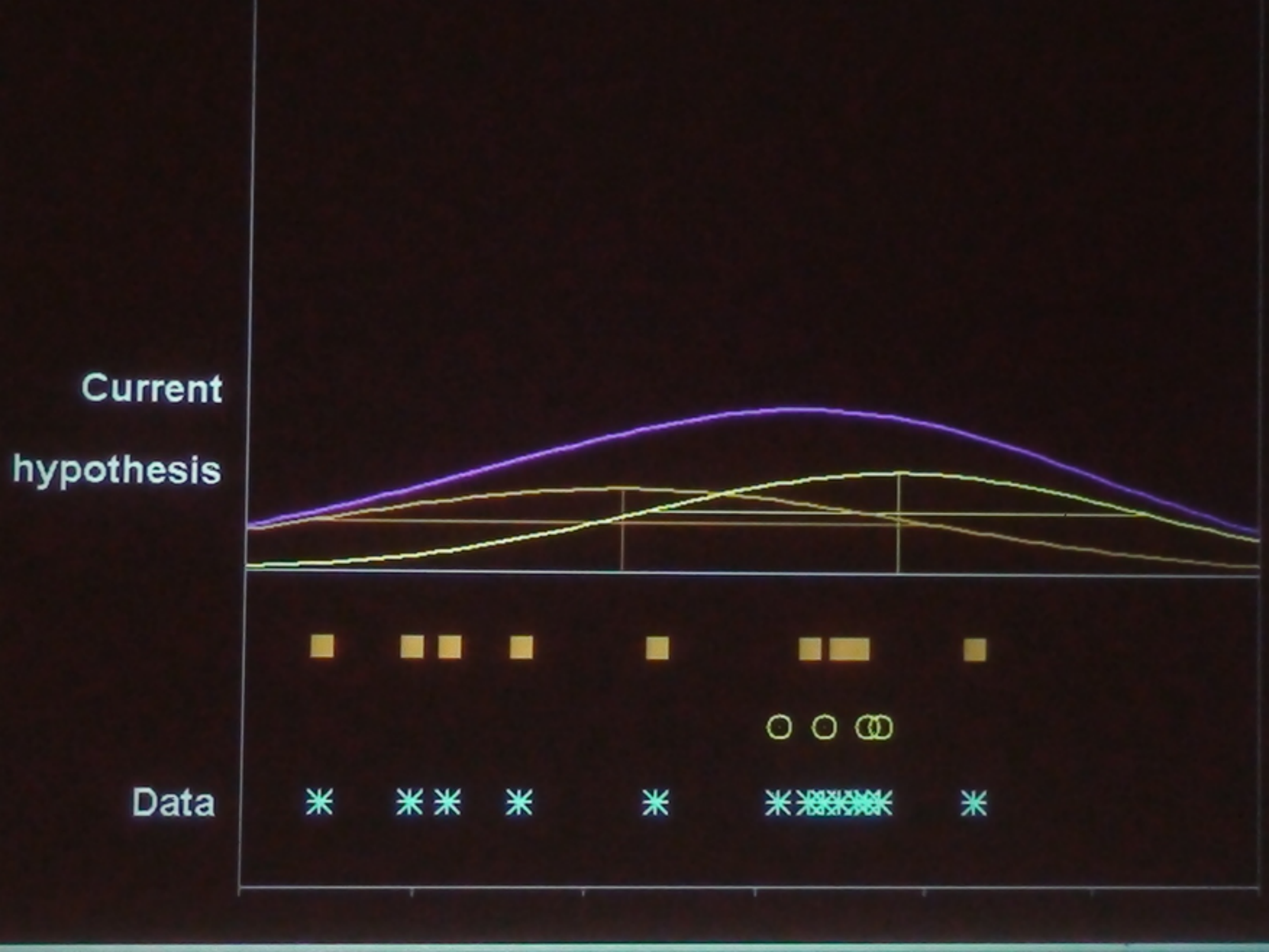
Clustering

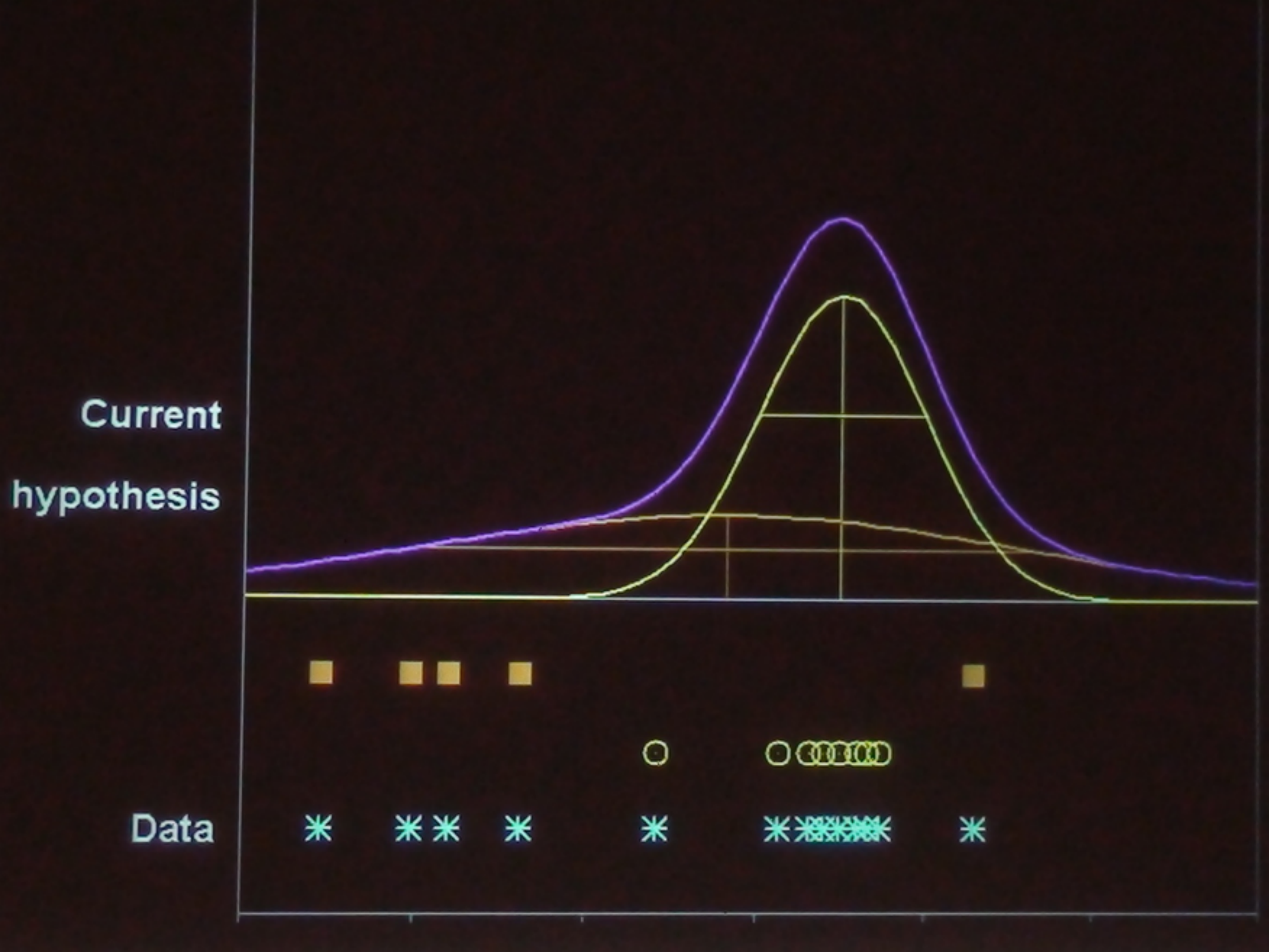
- K means clustering
  - Assign
  - Update
- Soft K means clustering
  - Assign
  - Update

Clustering

- K means clustering
  - Assign
  - Update
- Soft K means clustering
  - Assign
  - Update
- Clustering by Gibbs sampling
  - Assign
  - Update


Current
hypothesis
Data


Current
hypothesis
Data

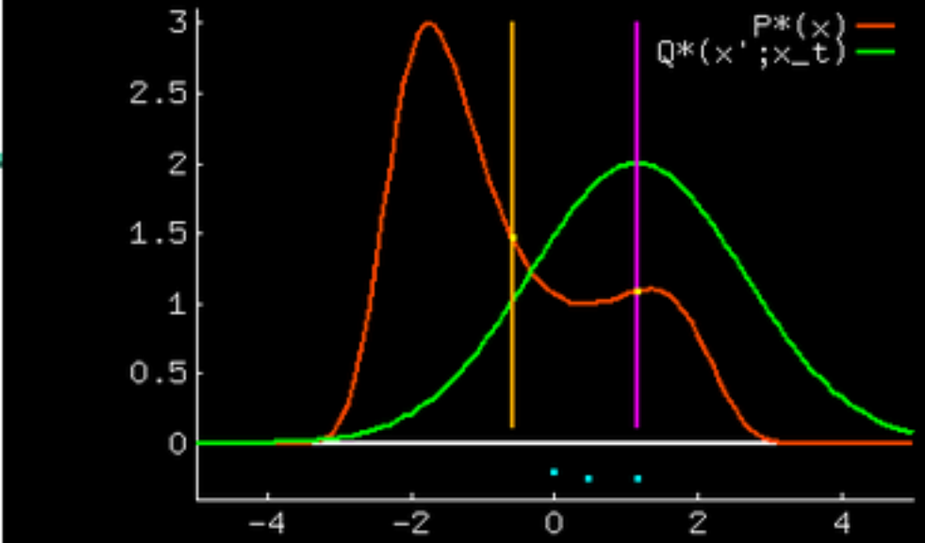

Current
hypothesis
Data

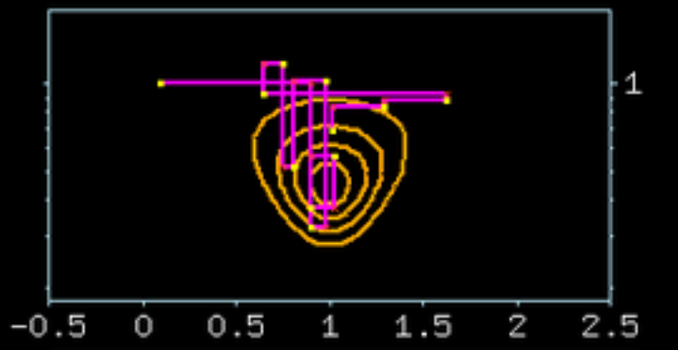
Monte Carlo methods

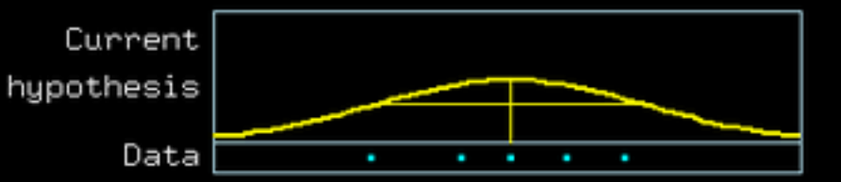
- Simple Monte Carlo methods
  - Importance sampling
  - Rejection sampling
- Markov-chain Monte Carlo methods
  - Metropolis method
  - Gibbs sampling

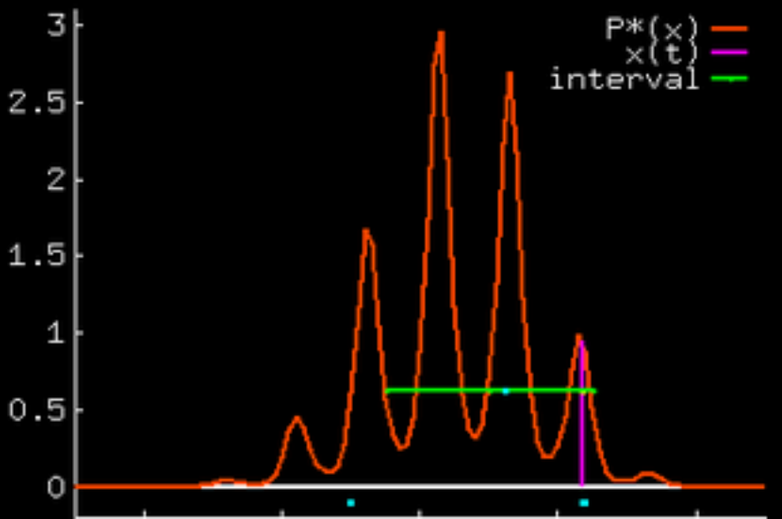
  - Slice sampling
- Reducing random-walk behaviour
  - Hamiltonian Monte Carlo
  - Overrelaxation
- Exact sampling

itp/exact/rc RUNME









Problems with standard Monte Carlo methods

- Random walk behaviour
- Sensitivity to step size
- When to stop

Efficient methods

Slice sampling

Exact sampling